PROBABILITY, STATISTICS AND RELIABILITY

[MAT3003]

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This material has been prepared in accordance with the syllabus of *Probability, Statistics and Reliability (MAT3003)* offered to B.Tech. and Integrated M.Tech. students at **VIT Bhopal University**. The content has been developed by **Dr. Sonjoy Pan** (Chapters 1 and 2), **Dr. Bhumika Choksi** (Chapter 3), **Dr. Mayank Sharma** (Chapter 4) and **Dr. Vinod Kumar Jatav** (Chapter 5).

Chapter 1

Probability Theory

- Fundamentals of Combinatorics: Counting Principle, Permutation, and Combination
- Introduction to Probability: Random Experiments, Sample Space, and Events and Its Various Types
- Definitions of Probability: Classical Probability, and Axiomatic Probability
- Conditional Probability: Total Probability, and Bayes' Theorem

Learning objectives

- Apply combinatorial methods (counting principles, permutations, combinations) to determine possible outcomes.
- Understand random experiments, sample spaces, and events, and compute probabilities of simple and compound events.
- Analyze event relationships using conditional probability and distinguish between independent and dependent events.
- Apply theorem of total probability and Bayes' theorem to solve complex problems and update probabilities based on new information.

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[Chapter 1 has been prepared by Dr. Sonjoy Pan]

1.1 Fundamentals of Combinatorics

Before studying probability theory, it is important to understand the basics of combinatorics, which deals with counting techniques and arrangements. Combinatorics provides methods for determining the number of possible outcomes in an experiment without listing them individually. Concepts such as permutations, combinations, and counting principles are fundamental tools in probability, as they help calculate the likelihood of events occurring. Thus, combinatorics forms an essential mathematical foundation for probability theory.

1.1.1 Counting Principle

Suppose that r experiments are to be performed such that the first one may result in any of n_1 possible outcomes, and for each of these n_1 possible outcomes, there are n_2 possible outcomes of the second experiment, and for each of the possible outcomes of the first two experiments, there are n_3 possible outcomes of the third experiment, and so on, then the total number of possible outcomes of the r experiments is $n_1 \cdot n_2 \cdot n_3 \cdots n_r$.

1.1.2 Permutations

The number of permutations (arrangements where order matters) of n distinct objects taken r at a time is represented as ${}_nP_r$ or ${}^nP_r$ which equals $\frac{n!}{(n-r)!}$.

The number ways (permutations) that n distinct objects to be placed in n positions (in line) is denoted by $n!$ which equals $n(n-1)(n-2)\cdots 2 \cdot 1$.

The number of different permutations of n objects, of which n_1 are alike, n_2 are alike, $\cdots$, n_r are alike is $\dfrac{n!}{n_1!n_2!\cdots n_r!}$.

1.1.3 Combinations

The number of combinations (groups where order does not matter) of n distinct objects taken r at a time is represented as $\binom{n}{r}$ or ${}_nC_r$ or ${}^nC_r$ which equals $\frac{n!}{r!(n-r)!}$.

The number of possible divisions of n distinct objects to be divided into r distinct groups of respective sizes n_1, n_2, $\cdots$, n_r, where $n_1 + n_2 + \cdots + n_r = n$ is represented as $\binom{n}{n_1, n_2, \cdots, n_r}$ which equals $\dfrac{n!}{n_1!n_2!\cdots n_r!}$.

1.2 Introduction to Probability

Probability is a branch of mathematics that deals with the study of uncertainty and random phenomena. It provides a systematic way to measure the chance of occurrence of different events. In everyday life, probability helps us analyze situations involving uncertainty, such as weather forecasting, games of chance, medical diagnosis, finance, engineering, and scientific experiments.

The concept of probability originated from the study of gambling problems and later developed into an important branch of mathematics and statistics. The value of probability always lies between 0 and 1, where 0 represents an impossible event and 1 represents a certain event.

To understand probability theory, it is necessary to study some fundamental concepts such as random experiments, sample spaces, events, and the mathematical definition of probability. These concepts form the foundation for further study in probability and statistics.

1.2.1 Random Experiments

If in an experiment conducted under identical conditions, the precise result is not unique, but may be any one of the possible results, then such an experiment is called a random experiment. Examples: tossing a coin, rolling a die, drawing a card from a pack of plying cards, etc.

1.2.2 Sample Space

The set of all possible outcomes of a random experiment is known as the sample space of the experiment and is denoted by S. Examples: {head, tail} and $\{1, 2, 3, 4, 5, 6\}$ are sample spaces of the respective random experiments.

1.2.3 Events and Its Various Types

In probability theory, an event is a collection of outcomes of a random experiment. Whenever an experiment is performed, different results may occur, and these results are called outcomes. Events help us describe and study these outcomes in a systematic way. For example, when a die is rolled, getting an even number is an event. Events can be classified into different types depending on their nature and relationships with other events.

Simple Events and Compound or Composite Events: An event is called simple if it contains exactly one outcome; otherwise, it is known as compound or composite. Examples: {tail} is simple event, whereas $\{2, 4\}$ is a compound event.

Sure Events or Certain Events: An event which always occurs is called sure event and is denoted by S. Examples: {head, tail} is a sure event in an experiment of tossing a coin.

Impossible Events or Null Events: An event which never occurs is called impossible event and is denoted by ϕ. Examples: $\{8\}$ is an impossible event in a random experiment of rolling a die.

Complementary Events: Two events are said to be complementary events if they are mutually exclusive and one of them must occur. The complementary event of an event A in sample space S is defined as $A^c = S - A$. Examples: {head} and {tail} are complementary events.

Mutually Exclusive Events or Disjoint Events: Two or more events are called mutually exclusive if any two of them cannot occur simultaneously. The events A and B are mutually exclusive if and only if $A \cap B = \phi$. Examples: the events $\{4, 6\}$ and $\{2, 3, 5\}$ are mutually exclusive.

Mutually Independent Events: Two or more events are called mutually independent if the occurrence of any event does not depend on the occurrence of other events. Examples: if we flip a coin and the outcome of this experiment is denoted by A; then again if we flip the coin and the outcome of this trail is denoted by B. Here A and B are independent events.

Exhaustive Events: The events $A_1, A_2, A_3, \cdots, A_n$ are said to be exhaustive events of sample space S if $\cup_{i=1}^{n} A_i = S$ and $A_i \cap A_j = \phi$ for $i, j = 1, 2, \cdots, n$ with $i \neq j$. Examples: the events {head} and {tail} are exhaustive in case of coin tossing; the events $\{1, 2\}$, $\{3, 5\}$ and $\{4, 6\}$ are also exhaustive in case of dice rolling.

Equally Likely Events: Several events are said to be equally likely if they have the equal chance to occur. Examples: $\{1\}$, $\{3\}$ and $\{6\}$ are equally likely events if we roll a die.

1.3 Definitions of Probability

Over time, different approaches have been developed to define probability more precisely. Among them, the classical definition and the axiomatic definition are two important foundations in the study of probability theory.

1.3.1 Classical Probability (Theoretical)

If all outcomes of a random experiment are equally likely to occur, then the probability of event A is expressed as

$$P(A) = \frac{\text{Number of outcomes in } A}{\text{Number of outcomes in } S}.$$

1.3.2 Axiomatic Probability (Modern Approach)

The probability of an event A in a sample space S, denoted by $P(A)$, is a number satisfying the following three axioms:

1. $0 \leq P(A) \leq 1$.
2. $P(\phi) = 0$ and $P(S) = 1$.
3. $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$, if $\{A_1, A_2, A_3, \cdots\}$ is a sequence of mutually exclusive events.

□ **Addition Rules:** If A and B are two events in a sample space, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

If A, B and C are three events in a sample space, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C).$$

In general, suppose that $A_1, A_2, \cdots, A_n$ are n events in some sample space, then

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i) - \sum_{\substack{i=1 \\ i<j}}^{n} \sum_{j=1}^{n} P(A_i \cap A_j) + \sum_{\substack{i=1 \\ i<j<k}}^{n} \sum_{j=1}^{n} \sum_{k=1}^{n} P(A_i \cap A_j \cap A_k) + \cdots + (-1)^{n+1} P(A_1 \cap \cdots \cap A_n).$$

☐ Some Useful Properties:

1. $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.
2. $(A^c)^c = A$ and $A \cap B^c = A - A \cap B$.
3. $P(A) + P(A^c) = 1$.
4. $P(A \cap B) = 0$, if and only if A and B are mutually exclusive (disjoint) events.
5. $P(A \cap B) = P(A)P(B)$, if and only if A and B are mutually independent events.
6. $P(A \cup B)$ *i.e.,* $P(A$ or $B)$ represents the probability that at least one of the events A or B occurs.
7. $P(A \cap B)$ *i.e.,* $P(A$ and $B)$ represents the probability that both events A and B occur together.

SOLVED PROBLEMS - 1A

Problem 1.1. *If* $P(A) = 0.7, P(B^c) = 0.6, P(A^c \cap B) = 0.3$, *find* $P(A \cap B^c)$.

Solution.
$$\begin{aligned} P(A \cap B^c) = P(A^c \cup B)^c &= 1 - P(A^c \cup B) \\ &= 1 - [P(A^c) + P(B) - P(A^c \cap B)] \\ &= 1 - [\{1 - P(A)\} + \{1 - P(B^c)\} - P(A^c \cap B)] \\ &= 1 - [\{1 - 0.7\} + \{1 - 0.6\} - 0.3] = 0.6. \end{aligned}$$

Problem 1.2. *Suppose* X *and* Y *are two mutually exclusive events, and* $P(X^c) = 0.45$, $P(Y^c) = 0.85$, *find* $P(X^c \cap Y^c)$.

Solution. *Since* X *and* Y *are two mutually exclusive events, then* $P(X \cap Y) = 0$.
Now,
$$\begin{aligned} P(X^c \cap Y^c) = P(X \cup Y)^c &= 1 - P(X \cup Y) \\ &= 1 - [P(X) + P(Y) - P(X \cap Y)] \\ &= 1 - [\{1 - P(X^c)\} + \{1 - P(Y^c)\} - P(X \cap Y)] \\ &= 1 - [\{1 - 0.45\} + \{1 - 0.85\} - 0] = 0.3. \end{aligned}$$

Problem 1.3. *Given* $P(X^c) = 0.6, P(Y) = 0.7$ *and* $P(X \cup Y) = 0.82$. *Examine whether* X *and* Y *are independent events.*

Solution. $P(X) = 1 - P(X^c) = 1 - 0.6 = 0.4$.
Now, $P(X)P(Y) = 0.4 \times 0.7 = 0.28$.
We have $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$, *which gives*
$$\begin{aligned} P(X \cap Y) &= P(X) + P(Y) - P(X \cup Y) \\ &= 0.4 + 0.7 - 0.82 = 0.28. \end{aligned}$$
Therefore, $P(X \cap Y) = P(X)P(Y)$. *Hence,* X *and* Y *are independent events.*

Problem 1.4. *Two dice are rolled simultaneously. Find the probability of getting the sum of their faces to be* 7 *or* 9.

Solution. *Let the consider the event as*

$A \equiv$ *The sum of faces of two dice is* 7.

$B \equiv$ *The sum of faces of two dice is* 9.

Clearly $A = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$.

and $B = \{(3,6),(4,5),(5,4),(6,3)\}$.

Since each die has 6 *outcomes and two dice are rolled simultaneously, then the number of outcomes in the sample space is* $6 \times 6 = 36$. *The number of outcomes in* A *is* 6 *and that in* B *is* 4. *Also,* $A \cap B = \phi$. *Hence, the required probability is* $P(A \cup B) = P(A) + P(B) = \frac{6}{36} + \frac{4}{36} = \frac{5}{18}$.

Problem 1.5. *Four fair coins are tossed simultaneously. Find the probability that at least one head turns up.*

Solution. *Let the consider the event as*

$A \equiv$ *At least one head turns up.*

Then $A^c \equiv$ *No head turns up* $\equiv$ *{TTTT}.*

Since each coin has two outcomes (head and tail) and four coins are tossed simultaneously, the number of outcomes in the sample space is $2^4 = 16$. *The number of outcomes in* A^c *is* 1.

Hence, $P(A^c) = \dfrac{1}{16}$. *Thus the required probability is* $P(A) = 1 - P(A^c) = 1 - \dfrac{1}{16} = \dfrac{15}{16}$.

Problem 1.6. *A box contains* 120 *cards numbered from* 1 *to* 120. *One card is drawn at random. Find the probability of getting number in the card divisible by either* 4 *or* 6.

Solution. *Let us consider the events as*

$A \equiv$ *The numbers in the card divisible by* 4.

$B \equiv$ *The numbers in the card divisible by* 6.

Clearly, $A \cap B \equiv$ *The numbers in the card divisible by both* 4 *and* 6.

$A \cup B \equiv$ *The numbers in the card divisible by either* 4 *or* 6.

The number of elements in A *is* $\dfrac{120}{4} = 30$.

The number of elements in B *is* $\dfrac{120}{6} = 20$.

The number of elements in $A \cap B$ *is* $\dfrac{120}{\textit{l.c.m.}\{4,\ 6\}} = \dfrac{120}{12} = 10$.

Therefore, $P(A) = \dfrac{30}{120},\ P(B) = \dfrac{20}{120}$ *and* $P(A \cap B) = \dfrac{10}{120}$.

Hence, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \dfrac{30}{120} + \dfrac{20}{120} - \dfrac{10}{120} = \dfrac{1}{3}$.

Problem 1.7. *If* 5 *cards are randomly drawn from a deck of* 52 *playing cards, find the probability of getting* 2 *queens and* 3 *aces.*

Solution. *The number of ways that* 2 *queens can be chosen from* 4 *queens is* ${}^4C_2$.

The number of ways that 3 *aces can be chosen from* 4 *aces is* ${}^4C_3$.

The number of ways that 5 *cards can be chosen from* 52 *cards is* ${}^{52}C_5$.

Therefore, the required probability is $\dfrac{{}^4C_2 \cdot {}^4C_3}{{}^{52}C_5} = \dfrac{4!}{2! \times 2!} \cdot \dfrac{4!}{3! \times 1!} \cdot \dfrac{5! \times 47!}{52!} = \dfrac{1}{108290}$.

1.4 Conditional Probability

The probability of an event B occurring when it is known that some event A has already occurred is called the conditional probability of B given A, denoted by $P(B|A)$ and defined as

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \text{ provided } P(A) \neq 0.$$

Two events A and B are independent if and only if $P(B|A) = P(B)$ or $P(A|B) = P(A)$.

□ **Multiplication Rules:** If A and B are two events in a sample space such that $P(A) \neq 0$ and $P(B) \neq 0$, then

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B).$$

If A, B and C are three events in a sample space, then

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B).$$

In general, if, in an experiment, the events $A_1, A_2, \cdots, A_n$ can occur, then

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_n|A_1 \cap A_2 \cap \cdots \cap A_{n-1}).$$

⊖ **NOTE:** $P(A \cap B) = P(A)P(B)$, if and only if A and B are mutually independent events.

1.4.1 Total Probability

If $A_1, A_2, A_3, \cdots, A_n$ are exhaustive events of sample space S with $P(A_i) \neq 0$, $i = 1, 2, \cdots, n$ and B is any other event in S, then

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n).$$

1.4.2 Bayes' Theorem

If $A_1, A_2, A_3, \cdots, A_n$ are exhaustive events of sample space S with $P(A_i) \neq 0$, $i = 1, 2, \cdots, n$ and B is any other event in S such that $P(B) \neq 0$, then

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n)}, \ i = 1, 2, \cdots, n.$$

SOLVED PROBLEMS - 1B

Problem 1.8. *A box contains* 12 *bulbs,* 5 *of which are defective.* 2 *bulbs are drawn at random (a) with replacement, (b) without replacement. Find the probability that none of those* 2 *bulbs is defective in both cases (a) and (b).*

Solution. *Let us consider the events as*

$A_1 \equiv$ *Non-defective bulb is selected in the first drawn.*

$A_2 \equiv$ *Non-defective bulb is selected in the second drawn.*

(a) ***With replacement:*** *In case of with replacement, the bulb is replaced after the first*

drawn, that is, the situation before the second drawing is the same as at the beginning. Here, the events A_1 and A_2 are independent.
Clearly, $P(A_1) = \frac{7}{12}$, since out of 12 bulbs, $12 - 5 = 7$ are non-defective.
Similarly, $P(A_2) = \frac{7}{12}$.
Hence, the required probability is $P(A_1 \cap A_2) = P(A_1)P(A_2) = \frac{7}{12} \cdot \frac{7}{12} = \frac{49}{144}$.
*(b) **Without replacement:** In case of without replacement, the bulb is not replaced after the first drawn, that is, after the first drawn, there are 11 bulbs left, out of which 6 are non-defective.*
Therefore, $P(A_1) = \frac{7}{12}$ and $P(A_2|A_1) = \frac{6}{11}$.
Hence, the required probability is $P(A_1 \cap A_2) = P(A_1)P(A_2|A_1) = \frac{7}{12} \cdot \frac{6}{11} = \frac{7}{22}$.

Problem 1.9. *In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history, and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that*
(a) the student took mathematics or history;
(b) the student did not take either of these subjects;
(c) the student took history but not mathematics.

Solution. *Let us consider the events as*
$M \equiv$ *Students took mathematics.*
$H \equiv$ *Students took history.*
Given that $P(M) = \frac{54}{100}$, $P(H) = \frac{69}{100}$ and $P(M \cap H) = \frac{35}{100}$.
(a) The required probability is $$P(M \cup H) = P(M) + P(H) - P(M \cap H) = \frac{54}{100} + \frac{69}{100} - \frac{35}{100} = \frac{22}{25}.$$
(b) The required probability is $P(M \cup H)^c = 1 - P(M \cup H) = 1 - \frac{22}{25} = \frac{3}{25}$.
(c) The required probability is $P(H \cap M^c) = P(H) - P(M \cap H) = \frac{69}{100} - \frac{35}{100} = \frac{17}{50}$.

Problem 1.10. *The probability that a married man watches a certain television show is 0.4 and the probability that a married woman watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find the probability that*
(a) a married couple watches the show;
(b) a wife watches the show, given that her husband does;
(c) at least one member of a married couple will watch the show.

Solution. *Let us consider the events as*
$H \equiv$ *Husband watches a certain TV show.*
$W \equiv$ *Wife watches the same TV show.*
Given that $P(H) = 0.4$, $P(W) = 0.5$ and $P(H|W) = 0.7$.
(a) The required probability is $P(H \cap W) = P(W)P(H|W) = 05 \times 0.7 = 0.35$.
(b) The required probability is $P(W|H) = \frac{P(W \cap H)}{P(H)} = \frac{0.35}{0.4} = 0.875$.
(c) The required probability is $$P(H \cup W) = P(H) + P(W) - P(H \cap W) = 0.4 + 0.5 - 0.35 = 0.55.$$

Problem 1.11. *One bag contains* 4 *white balls and* 3 *black balls, and a second bag contains* 3 *white balls and* 5 *black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?*

Solution. *Let us consider the events as*

$B_1 \equiv$ *Black ball is drawn from the first bag.*

$W_1 \equiv$ *White ball is drawn from the first bag.*

$B_2 \equiv$ *Black ball is drawn from the second bag.*

The drawn ball from the first bag can be black or white. Therefore, the entire scenario can be considered as a union of two events $(B_1 \cap B_2)$ *and* $(W_1 \cap B_2)$ *which are mutually exclusive.*

Thus the required probability is

$$\begin{aligned} P[(B_1 \cap B_2) \cup (W_1 \cap B_2)] &= P(B_1 \cap B_2) + P(W_1 \cap B_2) \\ &= P(B_1)P(B_2|B_1) + P(W_1)P(B_2|W_1) \\ &= \frac{3}{7} \cdot \frac{6}{9} + \frac{4}{7} \cdot \frac{5}{9} = \frac{38}{63}. \end{aligned}$$

Problem 1.12. *In a certain assembly plant, three machines,* B_1, B_2, *and* B_3, *make* 30%, 45%, *and* 25%, *respectively, of the products. It is known from past experience that* 2%, 3%, *and* 2% *of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. Find the probability that*

(a) the selected product is defective.

(b) the selected product was made by machine B_3, *if it is found to be defective.*

Solution. *Let us consider the events as*

$B_1 \equiv$ *The product is made by machine* B_1.

$B_2 \equiv$ *The product is made by machine* B_2.

$B_3 \equiv$ *The product is made by machine* B_3.

$D \equiv$ *The product is defective.*

Given that $P(B_1) = \dfrac{30}{100},\ P(B_2) = \dfrac{45}{100},\ P(B_3) = \dfrac{25}{100},$

$$P(D|B_1) = \frac{2}{100},\ P(D|B_2) = \frac{3}{100},\ P(D|B_3) = \frac{2}{100}.$$

(a) The required probability is

$$P(D) = P(B_1)P(D|B_1) + P(B_2)P(D|B_2) + P(B_3)P(D|B_3) = 0.0245.$$

(b) Using Bayes' formula, the required probability is

$$P(B_3|D) = \frac{P(B_3)P(D|B_3)}{P(B_1)P(D|B_1) + P(B_2)P(D|B_2) + P(B_3)P(D|B_3)} = \frac{0.25 \times 0.02}{0.0245} = \frac{10}{49}.$$

Problem 1.13. *Three urns contains* 6 *red,* 4 *black;* 4 *red,* 6 *black;* 5 *red,* 5 *black balls respectively. One of the urns is selected at random and a ball is drawn from it. If the ball drawn is red, find the probability that it is drawn from the first urn.*

Solution. *Let us consider the events as*

$U_1 \equiv$ *The ball is drawn from* U_1.

$U_2 \equiv$ *The ball is drawn from* U_2.

$U_3 \equiv$ *The ball is drawn from* U_3.

$R \equiv$ *The ball is red.*

Given that $P(U_1) = P(U_2) = P(U_3) = \dfrac{1}{3},\ P(R|U_1) = \dfrac{6}{10},\ P(R|U_2) = \dfrac{4}{10},\ P(R|U_3) = \dfrac{5}{10}.$

Therefore, $P(U_1|R) = \dfrac{P(U_1)P(R|U_1)}{P(U_1)P(R|U_1) + P(U_2)P(R|U_2) + P(U_3)P(R|U_3)} = \dfrac{2}{5}.$

Problem 1.14. *Only* 1 *in* 1000 *adults is afflicted with a rare disease for which a diagnostic test has been developed. The test is such that when an individual actually has the disease, a positive result will occur* 99% *of the time, whereas an individual without the disease will show a positive test result only* 2% *of the time. If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?*

Solution. *Let us consider the events as*

$A_1 \equiv$ *Individual has the disease.*

$A_2 \equiv$ *Individual does not have the disease.*

$B \equiv$ *Positive test result.*

Given that $P(A_1) = 0.001$, $P(A_2) = 0.999$, $P(B|A_1) = 0.99$, $P(B|A_2) = 0.02$.

The required probability is

$$P(A_1|B) = \frac{P(A_1 \cap B)}{P(B)} = \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2)}$$
$$= \frac{(0.001)(0.99)}{(0.001)(0.99) + (0.999)(0.02)} = 0.47.$$

Problem 1.15. *Two computers X and Y are to be marketed. A salesman who is assigned the job of finding customers for them has* 60% *and* 40% *chances respectively of succeeding in case of computer X and Y. The two computers can be sold independently. Given that he was able to sell at least one computer. What is the probability that computer X has been sold?*

Solution. *Let us consider the events as*

$X \equiv$ *Computer X is marketed.*

$Y \equiv$ *Computer Y is marketed.*

Given that $P(X) = \frac{60}{100} = 0.6$ *and* $P(Y) = \frac{40}{100} = 0.4$.

Now, $P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$
$= P(X) + P(Y) - P(X)P(Y)$, *since* X *and* Y *are independent.*
$= 0.6 + 0.4 - (0.6)(0.4) = 0.76$.

The required probability is $P[X|(X \cup Y)] = \frac{P[X \cap (X \cup Y)]}{P(X \cup Y)} = \frac{P(X)}{P(X \cup Y)} = \frac{0.6}{0.76} = 0.79$.

Problem 1.16. *A and B are two weak students of statistics and their chances of solving a problem in statistics correctly are* 30% *and* 20%*, respectively. The chance of their making a common error is* 0.5%*. If they solve a statistical problem independently and obtain the same answer, find the probability that their answer is correct.*

Solution. *Let us consider the events as*

$E_1 \equiv$ *Both A and B solve the problem correctly.*

$E_2 \equiv$ *Exactly one of them solves the problem correctly.*

$E_3 \equiv$ *Neither of them solves the problem correctly.*

$S \equiv$ *They get the same answer.*

Now, $P(E_1) = 0.3 \times 0.2 = 0.06$, $P(E_2) = 0.3 \times 0.8 + 0.7 \times 0.2 = 0.38$,
$P(E_3) = 0.7 \times 0.8 = 0.56$, $P(S|E_1) = 1$, $P(S|E_2) = 0$, $P(S|E_3) = 0.005$.

The required probability is $P(E_1|S) = \dfrac{P(E_1)P(S|E_1)}{P(E_1)P(S|E_1) + P(E_2)P(S|E_2) + P(E_3)P(S|E_3)} = 0.9554.$

EXERCISES

Ex 1.1. *7 balls are randomly withdrawn from an urn that contains 12 red, 16 blue, and 18 green balls. Find the probability that*
(a) 3 red, 2 blue, and 2 green balls are withdrawn;
(b) at least 2 red balls are withdrawn;
(c) all withdrawn balls are the same color;
(d) either exactly 3 red balls or exactly 3 blue balls are withdrawn.

Ex 1.2. *A and B throw alternatively a pair of dice. A wins if he throws 6 before B throws 7 and B wins if he throws 7 before A throws 6. Find their respective chances of winning, if A begins.*

Ex 1.3. *Consider two boxes, one containing 1 black and 1 white marble, the other 2 black and 1 white marble. A box is selected at random, and a marble is drawn from it at random. What is the probability that*
(a) the marble is black?
(b) the first box was the one selected given that the marble is white?

Ex 1.4. *A total of 36 members of a club play tennis, 28 play squash, and 18 play badminton. Furthermore, 22 of the members play both tennis and squash, 12 play both tennis and badminton, 9 play both squash and badminton, and 4 play all three sports. How many members of this club play at least one of three sports?*

Ex 1.5. *An urn contains 5 red, 6 blue, and 8 green balls. If a set of 3 balls is randomly selected, what is the probability that each of the balls will be*
(a) of the same color?
(b) of different colors?
Repeat under the assumption that the balls are sampled with replacement: whenever a ball is selected, its color is noted and it is replaced in the urn before the next selection.

Ex 1.6. *The chances that doctor X will diagnose a disease D correctly is 60%. The chances that a patient will die by his treatment after correct diagnosis is 20% and the chance of death by wrong diagnosis is 70%. A patient of doctor X, who had disease D, died. What is the probability that his disease was diagnosed correctly?*

Ex 1.7. *A person X speaks truth 7 out of 10 times. A die is tossed. He reports that there is a six. What is the probability that actually there was six?*

Ex 1.8. *In a class of 50 students, what is the probability that*
(a) all students will have different birthdays in a non-leap year?
(b) at least two students will have the same birthday in a non-leap year?

Ex 1.9. *Fifty books are placed at random in a shelf of a library. Find the probability that a particular set of three books will be*
(a) always together.
(b) never together.

Ex 1.10. *Five cards are randomly drawn from a well-shuffled pack of 52 cards. Find the probability that there are (out of drawn five cards)*

(a) two kings, one queen, one jack and one ace.
(b) three red and two black.
(c) two diamonds, two hearts and one spade.

Ex 1.11. *A couple has two children. What is the probability that both the children are boys, if it is known that?*
(a) younger child is a boy,
(b) older child is a boy,
(c) at least one of them is boy.

Ex 1.12. *An insurance company insured* 1000 *scooter drivers,* 3000 *car drivers and* 6000 *truck drivers. The probabilities that scooter, car and truck drivers meet an accident are* $0.02, 0.04, 0.25$ *respectively. One of the insured persons meets with an accident. What is the probability that he is a*
(a) car driver,
(b) truck driver.

Ex 1.13. *By examining the chest X-ray, probability that TB is detected when a person is actually suffering is* 0.99*. The probability that the doctor diagnoses incorrectly that a person has T.B. on the basis of X-ray is* 0.001*. In a certain city* 1 *in* 1000 *persons suffers from T.B. A person is selected at random is diagnosed to have T.B. What is the chance that he actually has T.B.?*

Ex 1.14. *A person goes to office either by car, scooter, bus or train. In the last few months, he used car* 10 *times, scooter* 30 *times, bus* 20 *times and train* 10 *times to reach the office. It is also found that he reached the office late* 3 *times,* 8 *times,* 2 *times and* 1 *times when he took scooter, bus, car and train, respectively.*
(a) What is the probability that he reached the office on time,
(b) Given that he reached the office on time, then what is the probability that he travelled by car?

Ex 1.15. *Two numbers are selected at random from integers* 1 *to* 9*.*
(a) If their sum is even, find the probability that (i) both the numbers are odd, (ii) one of these two numbers is 5*.*
(b) If it is known that one of those two numbers is 7*, find the probability that (i) their sum is odd, (ii) their sum is even.*

Ex 1.16. *A person X speaks the truth in* 80% *cases and another person Y speaks the truth in* 90% *cases. Find the probability that they contradict each other in stating the same fact.*

Ex 1.17. *At a party,* n *men take off their hats. The hats are then mixed up, and each man randomly selects one. We say that a match occurs if a man selects his own hat. What is the probability of*
(a) no matches?
(b) exactly k *matches?*

ANSWERS OF EXERCISES

Ex 1.1

$$\text{(a) } P = \frac{\binom{12}{3}\binom{16}{2}\binom{18}{2}}{\binom{46}{7}}, \qquad \text{(b) } P = 1 - \frac{\binom{34}{7} + \binom{12}{1}\binom{34}{6}}{\binom{46}{7}}$$

$$\text{(c) } P = \frac{\binom{12}{7} + \binom{16}{7} + \binom{18}{7}}{\binom{46}{7}}, \qquad \text{(d) } P = \frac{\binom{12}{3}\binom{34}{4} + \binom{16}{3}\binom{30}{4}}{\binom{46}{7}}$$

Ex 1.2

$$p_A = \frac{5}{11}, \quad p_B = \frac{6}{11}$$

Ex 1.3

$$\text{(a) } P(\text{black}) = \frac{7}{12}, \quad \text{(b) } P(\text{Box 1} \mid \text{white}) = \frac{3}{7}$$

Ex 1.4

$$n(\text{at least one sport}) = 43$$

Ex 1.5

Without replacement:

$$\text{(a) } P(\text{same color}) = \frac{\binom{5}{3} + \binom{6}{3} + \binom{8}{3}}{\binom{19}{3}}, \quad \text{(b) } P(\text{different colors}) = \frac{5 \cdot 6 \cdot 8}{\binom{19}{3}}$$

With replacement:

$$\text{(a) } P(\text{same color}) = \left(\frac{5}{19}\right)^3 + \left(\frac{6}{19}\right)^3 + \left(\frac{8}{19}\right)^3, \quad \text{(b) } P(\text{different colors}) = 3! \cdot \frac{5}{19} \cdot \frac{6}{19} \cdot \frac{8}{19}$$

Ex 1.6

$$P(\text{diagnosed correctly} \mid \text{died}) = \frac{3}{10}$$

Ex 1.7

$$P(\text{actually six} \mid \text{reports six}) = \frac{7}{22}$$

Ex 1.8

$$\text{(a) } P(\text{all different birthdays}) = \frac{365 \cdot 364 \cdot \cdots \cdot 316}{365^{50}}$$

$$\text{(b) } P(\text{at least two same birthday}) = 1 - \frac{365 \cdot 364 \cdot \cdots \cdot 316}{365^{50}}$$

Ex 1.9

$$\text{(a) } P(\text{three books together}) = \frac{48! \cdot 3!}{50!}$$

$$\text{(b) } P(\text{three books never together}) = 1 - \frac{48! \cdot 3!}{50!}$$

Ex 1.10

$$\text{(a)} \quad P = \frac{\binom{4}{2}\binom{4}{1}\binom{4}{1}\binom{4}{1}}{\binom{52}{5}}, \qquad \text{(b)} \quad P = \frac{\binom{26}{3}\binom{26}{2}}{\binom{52}{5}}, \qquad \text{(c)} \quad P = \frac{\binom{13}{2}\binom{13}{2}\binom{13}{1}}{\binom{52}{5}}$$

Ex 1.11

$$\text{(a)} \quad P(\text{both boys}) = \frac{1}{2}, \qquad \text{(b)} \quad P(\text{both boys}) = \frac{1}{2}, \qquad \text{(c)} \quad P(\text{both boys}) = \frac{1}{3}$$

Ex 1.12

$$P(\text{accident}) = 0.164, \quad \text{(a) } P(\text{car driver} \mid \text{accident}) = \frac{3}{41}, \quad \text{(b) } P(\text{truck driver} \mid \text{accident}) = \frac{75}{82}$$

Ex 1.13

$$P(\text{TB} \mid \text{diagnosed TB}) = \frac{0.99}{1.989} \approx 0.4977$$

Ex 1.14

$$\text{(a)} \quad P(\text{on time}) = \frac{4}{5}, \qquad \text{(b)} \quad P(\text{car} \mid \text{on time}) = \frac{1}{7}$$

Ex 1.15

$$(a) \quad \text{(i)} \quad P(\text{both odd} \mid \text{sum even}) = \frac{5}{8}, \qquad \text{(ii)} \quad P(\text{one number is } 5 \mid \text{sum even}) = \frac{1}{4}$$

$$(b) \quad \text{(i)} \quad P(\text{sum odd}) = \frac{1}{2}, \qquad \text{(ii)} \quad P(\text{sum even}) = \frac{1}{2}$$

Ex 1.16

$$P(\text{contradiction}) = 0.26$$

Ex 1.17

(a) $P(\text{no matches}) = \frac{!n}{n!}$, (b) $P(\text{exactly } k \text{ matches}) = \frac{\binom{n}{k}!(n-k)}{n!}$, where $!m$ denotes the number of derangements of m objects. That is,

$$!m = m!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^m \frac{1}{m!}\right) = \sum_{k=0}^{m} (-1)^k \frac{m!}{k!}.$$

Chapter 2

One-Dimensional Random Variables and Probability Distributions

- Introduction to Random Variables: Discrete Random Variables, and Continuous Random Variables
- Probability Distribution: Probability Mass Functions, Probability Density Functions, and Cumulative Distribution Functions
- Measures of Central Tendency and Dispersion: Expectation, Variance, and Standard Deviation
- Special Distributions: Bernoulli Distribution, Binomial Distribution, Poisson Distributions, Uniform Distribution, and Normal Distribution
- Measures and Functions of Probability Distributions: Moments, and Moment Generating Functions
- Statistical Tables: Cumulative Binomial Distribution, Cumulative Poisson Distribution, and Cumulative Standard Normal Distribution

Learning objectives

- Understand the concept of random variables and distinguish between discrete and continuous types.
- Define and interpret probability functions (PMF/PDF) and cumulative distribution functions (CDF) for different random variables.
- Compute and analyze expectation, variance, and moment generating functions (up to second moments).
- Apply and interpret standard probability distributions (Bernoulli, Binomial, Poisson, Uniform, Normal) in solving practical problems.

Module 02, MAT3003//Probability, Statistics and Reliability
Mathematics Division//School of Advanced Sciences and Languages// VIT Bhopal University
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2.1 Introduction to Random Variables

A random variable is a rule that assigns each outcome of a random experiment to a real number. For example, we consider an experiment of tossing two coins simultaneously and let X denotes the number of heads appeared. Then the outcomes and the corresponding values of X are listed as:

Outcomes	:	HH	HT	TH	TT
Values of X :		2	1	1	0

Here X can be treated as a random variable that associates real numbers $0,\ 1$ or 2 with the outcomes in the sample space.

Random Variable: A random variable (r.v.) is a real-valued function defined on the sample space of an experiment. It is defined as

$$X \text{ (random variable)} :\ S \text{ (sample space)} \to \mathbb{R} \text{ (set of real numbers)}.$$

□ Properties:

1. If X is random variables and c is a constant, then cX, $\frac{1}{X}$, $|X|$, $f(X)$ are also random variables.

2. If X and Y are random variables, and c and d are constants, then $cX + dY$, XY, $f(X, Y)$ are also random variables.

2.1.1 Discrete Random Variables

It is a random variable whose range is a countable set. A countable set can be either a finite set or a countably infinite set.

Example 1. Let X be a random variable denoting the number of heads in an experiment of tossing two coins. X is a discrete variable as its range is a finite set {0, 1, 2}.

2.1.2 Continuous Random Variables

It has a range in the forms of some interval (bounded or unbounded) of the real line.

Example 2. Let X be a random variable denoting the height of different people in a given region. Since the people can have different measures of height (not limited to just natural numbers or any countable set), X is a continuous random variable.

2.2 Probability Distribution

The probability distribution of a random variable can be determined by calculating the probability of occurrence of every value in the range of the random variable.

2.2.1 Probability Mass Functions

For discrete random variables, the term “probability mass function (pmf)" is used to describe their distributions.

If X is a discrete random variable with distinct values $x_1,\ x_2, \ldots, x_n, \ldots$, then the probability mass function of X is denoted by $p(x)$ and is defined as

$$p(x) = \begin{cases} P(X = x_i), & \text{if } x = x_i, \\ 0, & \text{if } x \neq x_i,\ i = 1, 2, \ldots \end{cases}$$

The set of ordered pairs $\{(x_i, p(x_i)) : i = 1, 2, \ldots\}$ specifies the probability distribution of X.

□ Properties:

1. $p(x_i) \geq 0,\ \forall\ i.$
2. $\sum_i p(x_i) = 1.$
3. $P(X = x_i) = p(x_i).$
4. $P\left(x_i \leq X \leq x_j\right) = \sum_{x=x_i}^{x_j} p(x).$
5. $P\left(X \in S_1 \mid X \in S_2\right) = \dfrac{P\left(X \in S_1 \cap S_2\right)}{P(X \in S_2)}.$

2.2.2 Probability Density Functions

For continuous random variables, the term “probability density function (pdf)" is used to describe their distributions.

If X is a continuous random variable and there exists a non-negative function $f(x)$ defined for all $x \in (-\infty, \infty)$, such that for any set B of real numbers,

$$P(X \in B) = \int_B f(x)dx,$$

then $f(x)$ is called the probability density function of X. Thus, $P(X \in B)$ represents the area between the density curve $y = f(x)$ and the x-axis over the interval B.

□ Properties:

1. $f(x) \geq 0.$
2. $\int_{-\infty}^{\infty} f(x)dx = 1.$
3. $P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) = \int_a^b f(x)dx.$
4. $P(X = c) = P(c \leq X \leq c) = \int_c^c f(x)dx = 0.$
5. $P\left(X \in B_1 \mid X \in B_2\right) = \dfrac{P\left(X \in B_1 \cap B_2\right)}{P(X \in B_2)}.$

2.2.3 Cumulative Distribution Functions

The distribution function (df) or cumulative distribution function (cdf) of a random variable X (discrete or continuous) is denoted by $F(x)$ and is defined as

$$F(x) = P(X \leq x), \quad -\infty < x < \infty.$$

□ **Properties:**

1. $P(a < X \leq b) = F(b) - F(a)$.
2. $0 \leq F(x) \leq 1$ and $F(x) \leq F(y)$, if $x \leq y$.
3. $F(-\infty) = 0$ and $F(\infty) = 1$.
4. $F(x)$ is a non-decreasing function of x.

□ If X is a discrete random variable with probability mass function $p(x)$, then its cumulative distribution function is expressed as

$$F(x) = P(X \leq x) = \sum_{t \leq x} p(t), \text{ for } -\infty < x < \infty.$$

Example 3. Suppose N is a random variable that denotes the number appeared on the top face when a die is rolled. The probability distribution of N is given by the following table; and the probability mass function and cumulative distribution function of N are presented in Fig. 2.1.

Outcomes ($N = n$)	1	2	3	4	5	6
pmf ($f(n)$)	1/6	1/6	1/6	1/6	1/6	1/6
cdf ($F(n)$)	1/6	2/6	3/6	4/6	5/6	1

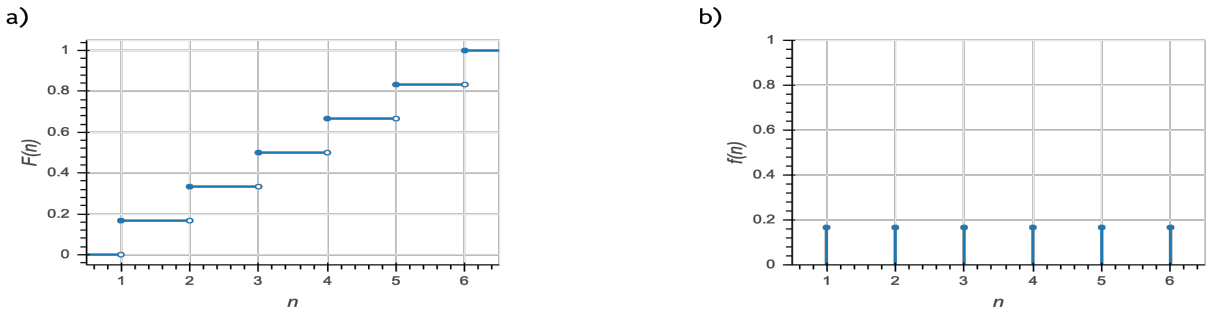


Figure 2.1: a) Discrete cumulative distribution function, b) Probability mass function, corresponding to Example 3.

□ If X is a continuos random variable with probability density function $f(x)$, then its cumulative distribution function is expressed as

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt, \text{ for } -\infty < x < \infty.$$

Here, the cdf $F(x)$ is a continuous function of x on the right and the discontinuities of $F(x)$ are at the most countable.
Therefore,

$$\frac{d}{dx}F(x) = f(x).$$

Theorem 1. Let X be a continuous random variable with probability density function $f_X(x)$ and $y = g(x)$ be strictly monotonic (increasing or decreasing) function of x. Assume that $g(x)$ is differentiable for all x. Then the probability density function $h_Y(y)$ of the random variable Y is given by

$$h_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|,$$

where x is expressed in terms of y and the range of Y is determined from the given range of X, on using the transformation $y = g(x)$.

Example 4. Consider a random variable Y that denotes the heights of men in centimeters in a given country. The probability density function and cumulative distribution function of Y are presented in Fig. 2.2.

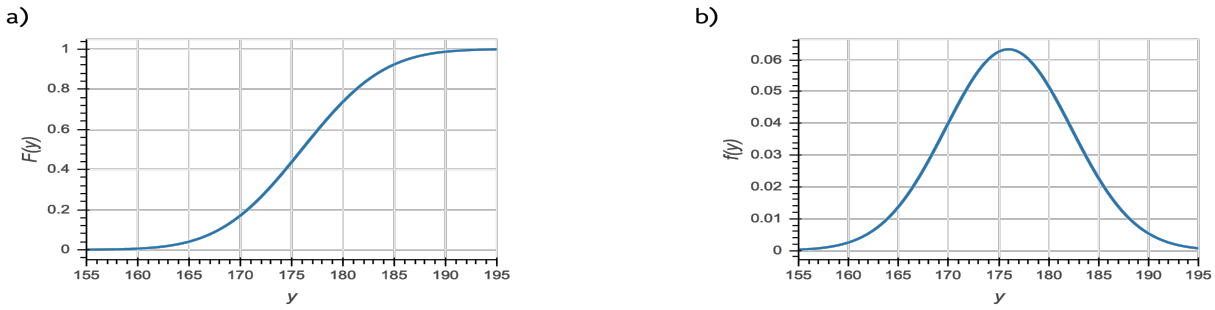


Figure 2.2: a) Continuous cumulative distribution function, b) Probability density function, corresponding to Example 4.

2.3 Measures of Central Tendency and Dispersion

In probability and statistics, numerical measures are used to describe the behavior and characteristics of a random variable or a data set. These measures help us understand the average value of observations as well as the extent to which the observations vary from one another. Such measures are very useful in analyzing data, comparing distributions, and making predictions.

Among these measures, Expectation, Variance, and Standard Deviation are the most important. Expectation gives the average or expected value of a random variable, while Variance and Standard Deviation measure the spread or dispersion of values around the average.

2.3.1 Expectation

Expectation of X: The expectation or mean of a discrete random variable X with probability mass function $p(x)$ is defined as

$$\mu_X = E(X) = \sum_x xp(x).$$

The expectation or mean of a continuous random variable X with probability density function $f(x)$ is defined as

$$\mu_X = E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

Expectation of g(X): The expectation of a function $g(X)$ of a discrete random variable X with probability mass function $p(x)$ is defined as

$$\mu_{g(X)} = E[g(X)] = \sum_x g(x)p(x).$$

The expectation of a function $g(X)$ of a continuous random variable X with probability density function $f(x)$ is defined as

$$\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx.$$

Properties:

- $E[c] = c, \;\; E[cX + d] = cE[X] + d.$
- $E[c_1X_1 + c_2X_2 + \cdots + c_nX_n] = c_1E[X_1] + c_2E[X_2] + \cdots + c_nE[X_n].$
- $E[X] \geq 0$, if $X \geq 0$.
- $E[X] \leq E[Y]$, if $X \leq Y$.

2.3.2 Variance

Variance of X: The variance of a random variable X is expressed as

$$\sigma_X^2 = \text{Var}(X) = E[X - E(X)]^2 = E[X^2] - (E[X])^2.$$

Thus,

$$\sigma_X^2 = \sum_x x^2p(x) - \mu_X^2, \;\; \text{for discrete random variable } X,$$

and

$$\sigma_X^2 = \int_{-\infty}^{\infty} x^2f(x)dx - \mu_X^2, \;\; \text{for continuous random variable } X.$$

Variance of $g(X)$: The variance of $g(X)$ is expressed as

$$\sigma_{g(X)}^2 = E[g(X) - \mu_{g(X)}]^2 = E[\{g(X)\}^2] - [\mu_{g(X)}]^2.$$

Therefore,

$$\sigma^2_{g(X)} = \sum_x \{g(x)\}^2 p(x) - \mu^2_{g(X)}, \quad \text{for discrete random variable } X,$$

and

$$\sigma^2_{g(X)} = \int_{-\infty}^{\infty} \{g(x)\}^2 f(x)dx - \mu^2_{g(X)}, \quad \text{for continuous random variable } X.$$

□ Properties:

- $\text{Var}(c) = 0$.
- $\text{Var}\,(cX + d) = c^2\text{Var}\,(X)$.

2.3.3 Standard Deviation

Standard Deviation of X: The standard deviation (SD) of a random variable X is defined as

$$\sigma_X = \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

SOLVED PROBLEMS - 2A

Problem 2.1. *A random variable X has the following probability function:*

$X = x$	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

(i) Find the value of k.
(ii) Evaluate $P(X < 6)$, $P(X \geq 6)$, $P(0 < X < 5)$.
(iii) Determine the cumulative distribution function of X.
(iv) Find the mean of X.
(v) Find the variance and standard deviation of X.

Solution. *(i)* $\sum_{x=0}^{7} p(x) = 1 \implies k = \frac{1}{10}, \; -1 \implies k = \frac{1}{10}$, *[since $p(x) \not< 0$].*

(ii) $P(X < 6) = \sum_{x=0}^{5} p(x) = 8k + k^2 = \frac{81}{100}$. $\quad P(X \geq 6) = 1 - P(X < 6) = \frac{19}{100}$.

$P(0 < X < 5) = \sum_{x=1}^{4} p(x) = 8k = \frac{4}{5}$.

(iii) The cumulative distribution function of X is given by

$$F_X(x) = P(X \leq x) = \sum_{-\infty}^{x} p(x).$$

$X = x$	0	1	2	3	4	5	6	7
$F_X(x) = P(X \leq x)$	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{5}{10}$	$\frac{8}{10}$	$\frac{81}{10}$	$\frac{83}{10}$	1

Hence,

$$F_X(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{10}, & 1 \le x < 2 \\ \frac{3}{10}, & 2 \le x < 3 \\ \frac{5}{10}, & 3 \le x < 4 \\ \frac{8}{10}, & 4 \le x < 5 \\ \frac{81}{10}, & 5 \le x < 6 \\ \frac{83}{10}, & 6 \le x < 7 \\ 1, & x \ge 7 \end{cases}$$

(iv) The mean of X is

$$\begin{aligned} E(X) = \sum_{x=0}^{7} xp(x) &= 0.0 + 1.k + 2.2k + 3.2k + 4.3k + 5.k^2 + 6.2k^2 + 7(7k^2 + k) \\ &= 30k + 66k^2 = \frac{183}{50}. \end{aligned}$$

(v) The variance of X is

$$\begin{aligned} Var(X) &= E(X^2) - [E(X)]^2 \\ &= \sum_{x=0}^{7} x^2 p(x) - \left(\frac{183}{50}\right)^2 \\ &= 0^2.0 + 1^2.k + 2^2.2k + 3^2.2k + 4^2.3k + 5^2.k^2 + 6^2.2k^2 + 7^2(7k^2 + k) - \left(\frac{183}{50}\right)^2 \\ &= 124k + 440k^2 - 13.3956 = 3.4044. \end{aligned}$$

The standard deviation of X is $\sqrt{3.4044} = 1.8451$.

Problem 2.2. *Let X be a continuous random variable with pdf:*

$$f(x) = \begin{cases} ax, & 0 \le x < 1 \\ a, & 1 \le x < 2 \\ -ax + 3a, & 2 \le x \le 3 \\ 0, & \textit{elsewhere} \end{cases}$$

(i) Determine the constant a.
(ii) Compute $P(X \le 1.5)$.
(iii) Determine the cumulative distribution function of X.
(iv) Find the mean of X.
(v) Find the variance and standard deviation of X.

Solution. *(i)* $\int_{-\infty}^{\infty} f(x)dx = 1 \implies \int_0^1 axdx + \int_1^2 adx + \int_2^3 (3a - ax)dx = 1 \implies a = \frac{1}{2}$.

(ii) $P(X \le 1.5) = \int_{-\infty}^{1.5} f(x)dx = \int_0^1 axdx + \int_1^{1.5} adx = a = \frac{1}{2}$.

(iii) The cumulative distribution function of X is given by

$$F_X(t) = P(X \leq t) = \int_{-\infty}^{t} f(x)dx.$$

If $t \leq 0$, $F_X(t) = 0$.
If $0 < t < 1$, $F_X(t) = \int_0^t f(x)dx = \int_0^t axdx = \frac{t^2}{4}$.
If $1 \leq t < 2$, $F_X(t) = \int_0^t f(x)dx = \int_0^1 axdx + \int_1^t adx = \frac{1}{4} + \frac{1}{2}(t-1) = \frac{t}{2} - \frac{1}{4}$.
If $2 \leq t < 3$, $F_X(t) = \int_0^t f(x)dx = \int_0^1 axdx + \int_1^2 adx + \int_2^t(-ax+3a)dx = -\frac{t^2}{4} + \frac{3t}{2} - \frac{5}{4}$.
If $t \geq 3$, $F_X(t) = 1$.
Hence,

$$F_X(t) = \begin{cases} 0, & t \leq 0 \\ \frac{t^2}{4}, & 0 < t < 1 \\ \frac{t}{2} - \frac{1}{4}, & 1 \leq t < 2 \\ -\frac{t^2}{4} + \frac{3t}{2} - \frac{5}{4}, & 2 \leq t < 3 \\ 1, & t \geq 3 \end{cases}$$

(iv) The mean of X is

$$\begin{aligned} E(X) = \int_{-\infty}^{\infty} xf(x)dx &= \int_0^1 x.axdx + \int_1^2 x.adx + \int_2^3 x(3a-ax)dx \\ &= a\left[\frac{x^3}{3}\right]_0^1 + a\left[\frac{x^2}{2}\right]_1^2 + a\left[\frac{3x^2}{2} - \frac{x^3}{3}\right]_2^3 \\ &= \frac{3}{2}. \end{aligned}$$

(v) The variance of X is

$$\begin{aligned} Var(X) &= E(X^2) - [E(X)]^2 \\ &= \int_{-\infty}^{\infty} x^2 f(x)dx - [E(X)]^2 \\ &= \int_0^1 x^2.axdx + \int_1^2 x^2.adx + \int_2^3 x^2(3a-ax)dx - [E(X)]^2 \\ &= a\left[\frac{x^4}{4}\right]_0^1 + a\left[\frac{x^3}{3}\right]_1^2 + a\left[x^3 - \frac{x^4}{4}\right]_2^3 - [E(X)]^2 \\ &= 0.4167. \end{aligned}$$

The standard deviation of X is $\sqrt{Var(X)} = 0.6455$.

Problem 2.3. *A shipment of 20 similar laptops to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these laptops, find the probability distribution for the number of defectives.*

Solution. *Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can be any of the numbers 0, 1, and 2. Now,*

$$f(0) = P(X=0) = \frac{\binom{3}{0}\binom{17}{2}}{\binom{20}{2}} = \frac{68}{95}, \quad f(1) = P(X=1) = \frac{\binom{3}{1}\binom{17}{1}}{\binom{20}{2}} = \frac{51}{190},$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2}\binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}.$$

Thus the probability distribution of X is

x	0	1	2
$f(x)$	$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$

Problem 2.4. *Let X be a continuous random variable with pdf*

$$f(x) = \begin{cases} \frac{3}{x^4}, & x \geq 1 \\ 0, & \textit{otherwise} \end{cases}$$

Find the mean and variance of X.

Solution.

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx = \int_1^{\infty} \frac{3}{x^3}dx = \lim_{n\to\infty}\left[-\frac{3}{2x^2}\right]_1^n = \frac{3}{2}.$$

Now,

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_1^{\infty} \frac{3}{x^2}dx = \lim_{n\to\infty}\left[-\frac{3}{x}\right]_1^n = 3.$$

Therefore, Var$(X) = E(X^2) - [E(X)]^2 = 3 - \frac{9}{4} = \frac{3}{4}$.

Problem 2.5. *A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.*

Solution. *Let X represent the number of good components in the sample. The probability distribution of X is*

$$f(x) = \frac{\binom{4}{x}\binom{3}{3-x}}{\binom{7}{3}}, \quad x = 0, 1, 2, 3.$$

Now,

$$f(0) = \frac{1}{35}, \quad f(1) = \frac{12}{35}, \quad f(2) = \frac{18}{35}, \quad f(3) = \frac{4}{35}.$$

Therefore,

$$E(X) = 0.f(0) + 1.f(1) + 2.f(2) + 3.f(3) = 0.\frac{1}{35} + 1.\frac{12}{35} + 2.\frac{18}{35} + 3.\frac{4}{35} = \frac{12}{5}.$$

Problem 2.6. *In a gambling game, a man is paid $\$5$ if he gets all heads or all tails when three coins are tossed simultaneously, and he will pay out $\$3$ if either one or two heads show. What is his expected gain?*

Solution. *The sample space for the possible outcomes when three coins are tossed simultaneously is*

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Let Y be the random variable that represents the amount the gambler can win. Therefore, the possible values of Y are given by

$$Y = \begin{cases} +\$5, & \text{if } E_1 = \{HHH, TTT\} \text{ occurs,} \\ -\$3, & \text{if } E_2 = \{HHT, HTH, THH, HTT, THT, TTH\} \text{ occurs.} \end{cases}$$

It is noted that $P(E_1) = 1/4$ and $P(E_2) = 3/4$. Therefore,

$$E(Y) = (5)(1/4) + (-3)(3/4) = -1.$$

Hence, in this game, the gambler will, on average, lose $\$1$ per toss of the three coins simultaneously.

2.4 Special Distributions

Different types of probability distributions are suitable for different kinds of random experiments and data. Among the most important and widely used probability distributions are the Bernoulli distribution, binomial distribution, Poisson distribution, uniform distribution, and normal distribution. The Bernoulli, binomial, and Poisson distributions deal with discrete random variables, whereas the uniform and normal distributions describe continuous random variables.

2.4.1 Bernoulli Distribution

A random variable X is said to have a Bernoulli distribution with parameter p if its probability mass function is

$$p(x) = \begin{cases} p, & \text{if } x = 1, \\ 1 - p, & \text{if } x = 0. \end{cases}$$

where $p \in [0, 1]$ represents the probability of success.

2.4.2 Binomial Distribution

A series of trials or an experiment is said to be a Bernoulli process if the following properties are satisfied:

1. The experiment consists of n repeated trials and each trial results in two outcomes, namely, success and failure.

2. The probability of success p remains constant in each trial and the repeated trials are independent.

The number X of successes in n Bernoulli trials is called a binomial random variable. The probability distribution of this discrete random variable is called the binomial distribution, and its values will be denoted by $b(x; n, p)$.

Suppose that n Bernoulli trials, each of which results in a success with probability p and a failure with probability $1 - p$. Then the probability distribution (probability

mass function) of the binomial random variable X, the number of successes in n independent trials, is

$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \cdots, n.$$

Theorem 2. The mean and variance of the binomial distribution $b(x; n, p)$ are

$$\mu = np \text{ and } \sigma^2 = np(1-p).$$

2.4.3 Poisson Distribution

An experiment is called Poisson process if it follows the following properties:

1. The number of outcomes occurring in one time interval or specified region is independent of the number that occurs in any other disjoint time interval or region of space.

2. The probability that a single outcome will occur during a very short time interval or in a small region is proportional to the length of the time interval or the size of the region and does not depend on the number of outcomes occurring outside this time interval or region.

3. The probability that more than one outcome will occur in such a short time interval or fall in such a small region is negligible.

The number X of outcomes occurring during a Poisson experiment is called a Poisson random variable, and its probability distribution is called the Poisson distribution.

The probability distribution of the Poisson random variable X, representing the number of outcomes occurring in a given time interval or specified region denoted by t, is

$$p(x; \mu) = \frac{e^{-\mu}\mu^x}{x!}, \quad x = 0, 1, 2, \cdots,$$

$\mu = \lambda t$, where λ is the average number of outcomes per unit time, distance, area, or volume.

Theorem 3. The mean and variance of the Poisson distribution $p(x; \mu)$ are equal and which is μ or λt.

Poisson Distribution as a Limiting Form of Binomial Distribution

Theorem 4. Let X be a binomial random variable with probability distribution $b(x; n, p)$. When $n \to \infty$, $p \to 0$ (or $p \to 1$), and $np \to \mu$ remains constant,

$$b(x; n, p) \to p(x; \mu).$$

2.4.4 Uniform Distribution

A random variable is said to be uniformly distributed on the interval (α, β) if its probability density function is given by

$$f(x;\alpha,\beta) = \begin{cases} \frac{1}{\beta-\alpha}, & \text{if } \alpha < x < \beta, \\ 0, & \text{elsewhere.} \end{cases}$$

The cumulative distribution function $F(x)$ of a uniform random variable X is given by

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x;\alpha,\beta)dx = \begin{cases} 0, & \text{if } x \leq \alpha, \\ \frac{x-\alpha}{\beta-\alpha}, & \text{if } \alpha < x < \beta, \\ 1, & \text{if } x \geq \beta. \end{cases}$$

Theorem 5. The mean and variance of the uniform random variable are $\frac{\alpha+\beta}{2}$ and $\frac{(\beta-\alpha)^2}{12}$, respectively.

2.4.5 Normal Distribution

A continuous random variable X having the bell-shaped distribution is called a normal random variable or normally distributed with parameters μ and σ if its probability density function is given by

$$n(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty.$$

A normal distribution is also known as a Gaussian distribution.

Theorem 6. The mean and variance of the normal random variable are μ and σ^2, respectively.

Standard Normal Distribution: If $\mu = 0$ and $\sigma = 1$, then the normal distribution is said to be standard normal distribution.

If X is a normal random variable with mean μ and variance σ^2, then $Z = \frac{X-\mu}{\sigma}$ is the standard normal distribution.

The cumulative distribution of a standard normal random variable Z is given by

$$F(z) = P(Z \leq z) = \Phi(z).$$

It is noted that

$$P(Z \leq -z) = \Phi(-z) = 1 - \Phi(z).$$

Moreover,

$$P(z_1 \leq Z \leq z_2) = \Phi(z_2) - \Phi(z_1).$$

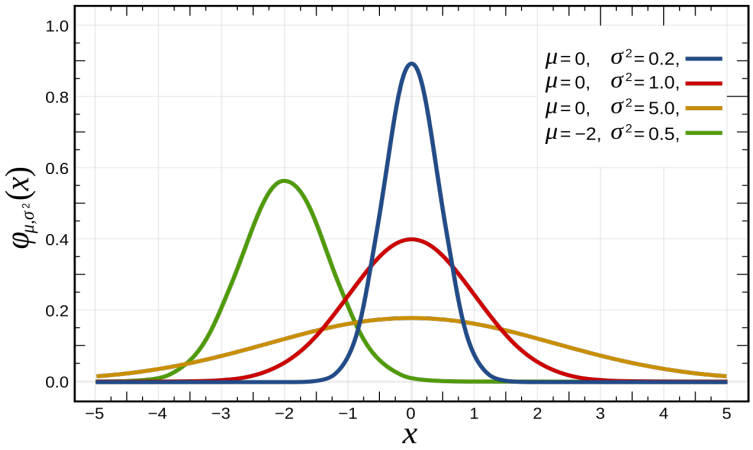


Figure 2.3: Normal distribution curves with different means and variances.

Theorem 7. Let $X_1,\ X_2, \cdots, X_n$ be n independent normal random variables with parameters (μ_1, σ_1), (μ_2, σ_2), $\cdots$, (μ_n, σ_n), respectively. Then $c_1X_1 + c_2X_2 + \cdots + c_nX_n$ is a normal random variable with mean $c_1\mu_1 + c_2\mu_2 + \cdots + c_n\mu_n$ and standard deviation $\sqrt{c_1^2\sigma_1^2 + c_2^2\sigma_2^2 + \cdots + c_n^2\sigma_n^2}$.

Normal Approximation to Binomial Distribution

Theorem 8. Let X be a binomial random variable with probability distribution $b(x; n, p)$. When $n \to \infty$, $p \to \frac{1}{2}$, $np \to \mu$, and $np(1-p) \to \sigma^2$ remains constant,

$$b(x; n, p) \to n(x; \mu, \sigma).$$

SOLVED PROBLEMS - 2B

Problem 2.7. *The probability that a patient recovers from a rare blood disease is* 0.4. *If* 15 *people are known to have contracted this disease, find the probability that*
(a) at least 10 *survive,*
(b) from 3 *to* 8 *survive,*
(c) exactly 5 *survive.*

Solution. *Let* X *be the number of people who survive. Then* X *is a binomial random variable.*

(a) $P(X \geq 10) = 1 - P(X < 10) = 1 - \sum_{x=0}^{9} b(x; 15, 0.4) = 1 - 0.9662 = 0.0338.$

(b) $P(3 \leq X \leq 8) = \sum_{x=3}^{8} b(x; 15, 0.4) = \sum_{x=0}^{8} b(x; 15, 0.4) - \sum_{x=0}^{2} b(x; 15, 0.4)$

$$= 0.9050 - 0.0271 = 0.8779.$$

(c) $P(X = 5) = b(5; 15, 0.4) = \sum_{x=0}^{5} b(x; 15, 0.4) - \sum_{x=0}^{4} b(x; 15, 0.4) = 0.4032 - 0.2173 = 0.1859.$

Problem 2.8. *The average number of oil tankers arriving each day at a certain port city is* 10*. The facilities at the port can handle at most* 15 *tankers per day. What is the probability that on a given day tankers have to be turned away, if the number of oil tankers arriving each day follows a Poisson distribution?*

Solution. *Let* X *be the number of tankers arriving each day. Then* $\mu = 10$*.*

$$P(X > 15) = 1 - P(X \leq 15) = 1 - \sum_{x=0}^{15} p(x; 10) = 1 - 0.9513 = 0.0487.$$

Problem 2.9. *In a certain industrial facility, accidents occur infrequently. It is known that the probability of an accident on any given day is* 0.005 *and accidents are independent of each other. Find the probability that*
(a) in any given period of 400 *days there will be an accident on one day.*
(b) there are at most three days with an accident.

Solution. *Let* X *be a binomial random variable with* $n = 400$ *and* $p = 0.005$*.*
Therefore, $\mu = np = 2$*. Using the Poisson approximation,*

(a) $P(X = 1) = p(1; 2) = \dfrac{e^{-2}2^1}{1!} = 0.271.$

(b) $P(X \leq 3) = \sum_{x=0}^{3} p(x; 2) = \sum_{x=0}^{3} \dfrac{e^{-2}2^x}{x!} = 0.857.$

Problem 2.10. *Let the random variable* X *be uniformly distributed over the interval* $[2, 8]$*. Find*
(a) the probability density function of X*.*
(b) $P(3 \leq X \leq 6)$*.*
(c) the mean and variance of X*.*

Solution. *For a uniform distribution* $X \sim U(a, b)$*, the probability density function is given by*

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \textit{otherwise} \end{cases}$$

Here, $a = 2$ *and* $b = 8$*.*

(a) The probability density function of X *is*

$$f(x) = \begin{cases} \frac{1}{8-2} = \frac{1}{6}, & 2 \leq x \leq 8 \\ 0, & \textit{otherwise.} \end{cases}$$

(b)

$$P(3 \leq X \leq 6) = \frac{6-3}{8-2} = \frac{3}{6} = \frac{1}{2}.$$

(c) Mean:

$$E(X) = \frac{2+8}{2} = 5.$$

Variance:

$$Var(X) = \frac{(8-2)^2}{12} = \frac{36}{12} = 3.$$

Problem 2.11. *A certain type of storage battery lasts, on average, 3 years with a standard deviation of 0.5 years. Assuming that the battery lives are normally distributed, find the probability that a given battery will last less than 2.3 years.*

Solution. *Let X represents the lifetime of a battery (in years). Then X is a normal random variable with mean $\mu = 3$ and standard deviation $\sigma = 0.5$.*
Therefore, $Z = \frac{X-\mu}{\sigma} = \frac{X-3}{0.5}$ represents the standard normal random variable. Thus, the required probability is

$$P(X < 2.3) = P\left(\frac{X-3}{0.5} < \frac{2.3-3}{0.5}\right) = P(Z < -1.4)$$

$$= \Phi(-1.4) = 1 - \Phi(1.4) = 1 - 0.9192 = 0.0808.$$

Problem 2.12. *An electrical firm manufactures light bulbs that have a life, before burnout, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns*
(a) between 778 and 834 hours.
(b) exactly 820 hours.

Solution. *Let X represents the lifetime of a bulb. Then X is a normal random variable with mean $\mu = 800$ and standard deviation $\sigma = 40$.*
Therefore, $Z = \frac{X-\mu}{\sigma} = \frac{X-800}{40}$ represents the standard normal random variable.
(a) The probability that a bulb burns between 778 and 834 hours is

$$P(778 < X < 834) = P\left(\frac{778-800}{40} < \frac{X-800}{40} < \frac{834-800}{40}\right)$$

$$= P(-0.55 < Z < 0.85) = P(Z < 0.85) - P(Z < -0.55)$$

$$= \Phi(0.85) - \Phi(-0.55) = \Phi(0.85) - [1 - \Phi(0.55)]$$

$$= 0.8023 - [1 - 0.7088] = 0.5111.$$

(b) The probability that a bulb burns exactly 790 hours is

$$P(X = 820) = 0.$$

2.5 Measures and Functions of Probability Distributions

In probability theory, it is often necessary to describe and analyze the properties of a probability distribution in a systematic way. This helps in understanding the overall behavior, shape, and characteristics of distributions. Moments help in describing different aspects of a distribution such as its central tendency, dispersion, skewness, and kurtosis, while moment generating functions provide a convenient method to generate these moments and study the properties of random variables.

2.5.1 Moments

The r-th moment of a random variable X about the point $X = a$ is denoted by μ'_r and defined as $\mu'_r = E\left[(X-a)^r\right]$.

The r-th moment of X about the origin is defined as

$$\mu'_r = E(X^r) = \begin{cases} \sum_x x^r p(x), & \text{if } x \text{ is discrete,} \\ \int_{-\infty}^{\infty} x^r f(x)dx, & \text{if } x \text{ is continuous.} \end{cases}$$

The first and second moments about the origin of a random variable X are given by

$$\mu'_1 = E(X) = \mu \quad \text{and} \quad \mu'_2 = E(X^2) = \mu^2 + \sigma^2.$$

2.5.2 Moment Generating Functions

The moment generating function (mgf) of a random variable X about the point $X = a$ is denoted by $M_X(t)$, t being the real parameter and defined as $M_X(t) = E\left[e^{t(X-a)}\right]$.

The moment generating function of X about the origin is defined as

$$M_X(t) = E(e^{tX}) = \begin{cases} \sum_x x^{tx} p(x), & \text{if } x \text{ is discrete,} \\ \int_{-\infty}^{\infty} x^{tx} f(x)dx, & \text{if } x \text{ is continuous.} \end{cases}$$

Now,

$$\begin{aligned} M_X(t) = E(e^{tX}) &= E\left[1 + (tX) + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \cdots + \frac{(tX)^r}{r!} + \cdots\right] \\ &= 1 + tE(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \cdots + \frac{t^r}{r!}E(X^r) + \cdots \\ &= 1 + t\mu'_1 + \frac{t^2}{2!}\mu'_2 + \frac{t^3}{3!}\mu'_3 + \cdots + \frac{t^r}{r!}\mu'_r + \cdots \\ &= \sum_{r=0}^{\infty} \frac{t^r}{r!}\mu'_r. \end{aligned}$$

Hence, the coefficient of $\frac{t^r}{r!}$ in the expression of moment generating function, $M_X(t)$, gives the r-th moment of X about the origin.

Theorem 9. Let X be a random variable with the r-th moment about the origin denoted by μ'_r and moment generating function denoted by $M_X(t)$. Then

$$\mu'_r = \left[\frac{d^r}{dt^r}\left\{M_X(t)\right\}\right]_{t=0}.$$

SOLVED PROBLEMS - 2C

Problem 2.13. *Let X be a continuous random variable with probability density function*

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & 0 \le x < \infty \\ 0, & \textit{elsewhere,} \end{cases}$$

where $\lambda > 0$. Obtain the moment generating function of X. Hence, find the mean and variance of X.

Solution. *The moment generating function of X is*

$$\begin{aligned} M_X(t) = E\left(e^{tX}\right) &= \int_{-\infty}^{\infty} e^{tx} f(x)dx \\ &= \int_0^{\infty} e^{tx}\lambda e^{-\lambda x}dx \\ &= \lambda\int_0^{\infty} e^{(t-\lambda)x}dx, \text{ which is finite only if } t<\lambda \\ &= \frac{\lambda}{t-\lambda}\lim_{n\to\infty}\left[e^{(t-\lambda)x}\right]_{x=0}^{n} \\ &= \frac{\lambda}{t-\lambda}(0-1), \quad t<\lambda \\ &= \frac{\lambda}{\lambda-t}, \quad t<\lambda. \end{aligned}$$

Now,

$$E(X) = \left[\frac{d}{dt}\left\{M_X(t)\right\}\right]_{t=0} = \left[\frac{\lambda}{(\lambda-t)^2}\right]_{t=0} = \frac{1}{\lambda}.$$

$$E(X^2) = \left[\frac{d^2}{dt^2}\left\{M_X(t)\right\}\right]_{t=0} = \left[\frac{2\lambda}{(\lambda-t)^3}\right]_{t=0} = \frac{2}{\lambda^2}.$$

Hence, $\mu_X = \frac{1}{\lambda}$ and $\sigma_X^2 = \frac{2}{\lambda^2} - \left[\frac{1}{\lambda}\right]^2 = \frac{1}{\lambda^2}$.

Problem 2.14. *Find the moment-generating function of a binomial random variable X. Further, find the first and second moments about the origin of X, and verify that $\mu = np$ and $\sigma^2 = np(1-p)$.*

Solution. *We have*

$$M_X(t) = \sum_{x=0}^{n} e^{tx}\binom{n}{x}p^x(1-p)^{n-x} = \sum_{x=0}^{n}\binom{n}{x}(pe^t)^x(1-p)^{n-x} = (pe^t+1-p)^n.$$

Now,

$$\frac{dM_X(t)}{dt} = n(pe^t+1-p)^{n-1}pe^t$$

and

$$\frac{d^2M_X(t)}{dt^2} = np[e^t(n-1)(pe^t+1-p)^{n-2}pe^t + (pe^t+1-p)^{n-1}e^t]$$

The first and second moments about the origin of X, respectively, are

$$\mu_1^{'} = \frac{dM_X(t)}{dt}\Big|_{t=0} = np,$$

$$\mu_2^{'} = \frac{d^2M_X(t)}{dt^2}\Big|_{t=0} = np[(n-1)p+1].$$

Therefore,

$$\mu = \mu_1^{'} = np,$$

$$\sigma^2 = \mu_2^{'} - \mu^2 = np(1-p).$$

Problem 2.15. *Show that the moment-generating function of the random variable X having a normal probability distribution with mean μ and variance σ^2 is given by*

$$M_X(t) = exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right).$$

Solution. *The moment-generating function of the normal random variable X is*

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}\sigma} exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} exp\left[-\frac{x^2 - 2(\mu + t\sigma^2)x + \mu^2}{2\sigma^2}\right] dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} exp\left[-\frac{\{x-(\mu+t\sigma^2)\}^2 - 2\mu t\sigma^2 - t^2\sigma^4}{2\sigma^2}\right] dx$$

$$= exp\left[\frac{2\mu t + t^2\sigma^2}{2}\right] \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} exp\left[-\frac{\{x-(\mu+t\sigma^2)\}^2}{2\sigma^2}\right] dx.$$

Let $z = \frac{x-(\mu+t\sigma^2)}{\sigma}$. Then $dx = \sigma dz$. Therefore,

$$M_X(t) = exp\left(\frac{2\mu t + t^2\sigma^2}{2}\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = exp\left(\frac{2\mu t + t^2\sigma^2}{2}\right).$$

As $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$ represents the area under a standard normal density curve, its value is equal to 1. Hence

$$M_X(t) = exp\left(\frac{2\mu t + t^2\sigma^2}{2}\right).$$

2.6 Statistical Tables

2.6.1 Table-I: Cumulative Binomial Distribution

For a binomial random variable $X \sim B(n, p)$,

$$P(X \leq x) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k}$$

The following table gives cumulative probabilities $P(X \leq x)$ for selected values of p when $n = 10$.

x	$p = 0.1$	$p = 0.2$	$p = 0.3$	$p = 0.4$	$p = 0.5$	$p = 0.6$
0	0.3487	0.1074	0.0282	0.0060	0.0010	0.0001
1	0.7361	0.3758	0.1493	0.0464	0.0107	0.0017
2	0.9298	0.6778	0.3828	0.1673	0.0547	0.0123
3	0.9872	0.8791	0.6496	0.3823	0.1719	0.0548
4	0.9984	0.9672	0.8497	0.6331	0.3769	0.1662
5	0.9999	0.9936	0.9527	0.8338	0.6230	0.3669
6	1.0000	0.9991	0.9894	0.9452	0.8281	0.6177
7	1.0000	0.9999	0.9984	0.9877	0.9453	0.8327
8	1.0000	1.0000	0.9999	0.9983	0.9893	0.9536
9	1.0000	1.0000	1.0000	0.9999	0.9990	0.9893
10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Note:

- Rows correspond to the number of successes x.
- Columns correspond to different probabilities of success p.
- Entries represent cumulative probabilities $P(X \leq x)$.
- This table is constructed for $n = 10$.

2.6.2 Table-II: Cumulative Poisson Distribution

For a Poisson random variable $X \sim \text{Poisson}(\lambda)$,

$$P(X \leq x) = \sum_{k=0}^{x} \frac{e^{-\lambda}\lambda^k}{k!}$$

The following table gives cumulative probabilities $P(X \leq x)$ for selected values of λ.

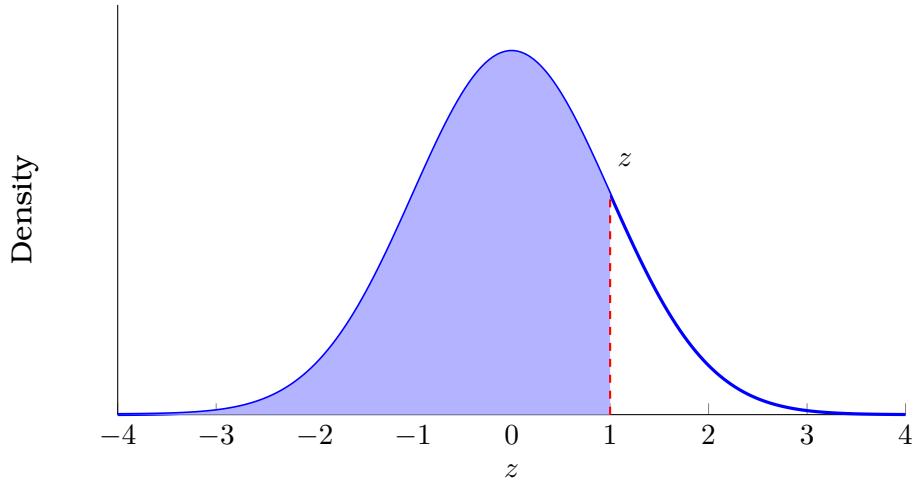
x	$\lambda = 0.5$	$\lambda = 1.0$	$\lambda = 1.5$	$\lambda = 2.0$	$\lambda = 2.5$	$\lambda = 3.0$
0	0.6065	0.3679	0.2231	0.1353	0.0821	0.0498
1	0.9098	0.7358	0.5578	0.4060	0.2873	0.1991
2	0.9856	0.9197	0.8088	0.6767	0.5438	0.4232
3	0.9982	0.9810	0.9344	0.8571	0.7576	0.6472
4	0.9998	0.9963	0.9814	0.9473	0.8912	0.8153
5	1.0000	0.9994	0.9955	0.9834	0.9580	0.9161
6	1.0000	0.9999	0.9989	0.9955	0.9858	0.9665
7	1.0000	1.0000	0.9998	0.9989	0.9967	0.9881
8	1.0000	1.0000	1.0000	0.9998	0.9993	0.9962
9	1.0000	1.0000	1.0000	1.0000	0.9998	0.9989
10	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997

Note:

- Rows correspond to values of the random variable x.
- Columns correspond to different Poisson means λ.
- Entries represent cumulative probabilities $P(X \leq x)$.

2.6.3 Table-IIIA: Cumulative Standard Normal Distribution

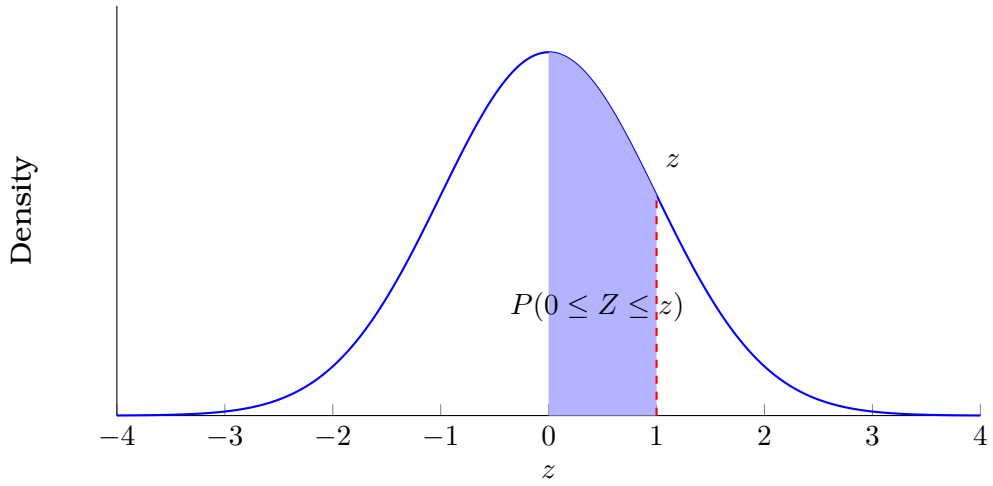
Values in the table represent the area under the standard normal curve to the left of z score, i.e., $P(Z \leq z)$ where $Z \sim N(0, 1)$.



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441

2.6.4 Table-IIIB: Cumulative Standard Normal Distribution (Different Form of Normal Curve)

Values in the table represent the area under the standard normal curve between 0 and z, i.e., $P(0 \leq Z \leq z)$ where $Z \sim N(0,1)$.



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441

EXERCISES

Probability Distribution and Distribution Function

Ex 2.1. *An important factor in solid missile fuel is the particle size distribution. Significant problems occur if the particle sizes are too large. From production data in the past, it has been determined that the particle size (in micrometers) distribution is characterized by*

$$f(x) = \begin{cases} 3x^{-4}, & x > 1 \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Verify that this is a valid density function
(b) Evaluate $F(x)$.
(c) What is the probability that a random particle from the manufactured fuel exceeds 4 micrometers?
(d) Obtain the probability density function of $Y = e^X$, where X denotes the particle size.

Ex 2.2. *The diameter of an electric cable, denoted by X, is assumed to be a continuous random variables with probability distribution: $f(x) = 6x(1-x), \quad 0 \leq x \leq 1$.*
(a) Check that $f(x)$ is a probability density function.
(b) Find the cumulative distribution function of X.
(c) Compute $P(X \leq 0.5 \mid 0.3 \leq X \leq 0.7)$.
(d) Determine the value of k such that $P(X < k) = P(X > k)$.

Ex 2.3. *A discrete random variable X has the distribution function:*

$$F(x) = \begin{cases} 0, & x < -2 \\ \frac{1}{2}, & -2 \leq x < 0 \\ \frac{3}{4}, & 0 \leq x < 2 \\ 1, & x \geq 2. \end{cases}$$

(a) Sketch the graph of $F(x)$.
(b) Obtain the probability mass function $f(x)$.
(c) Compute $P(X = 1)$, $P(X \leq 1)$, $P(X = 2)$ and $P(0 < X \leq 2)$.

Ex 2.4. *A random variable X assumes the values -3, -2, -1, 0, 1, 2 and 3 such that $P(X = -3) = P(X = -2) = P(X = -1)$, $P(X = 1) = P(X = 2) = P(X = 3)$ and $P(X < 0) = P(X = 0) = P(X > 0)$.*
(a) Find the probability mass function of X.
(b) Obtain distribution function of X.
(c) Find the probability mass function of $Y = 2X^2 + 3X + 4$.

Expectation and Variance

Ex 2.5. *The distribution function of a continuous random variable X is given by*

$$F(x) = \begin{cases} 0, & x \leq 1 \\ k(x-1)^4, & 1 < x \leq 3 \\ 1, & x > 3. \end{cases}$$

(a) Find k.
(b) Obtain the probability density function $f(x)$.
(c) Calculte the mean and variance of X.
(d) Compute $P(0.5 < X \leq 2.5)$.

Ex 2.6. *The following is the distribution function of a discrete random variable X:*

x	-3	-1	0	1	2	3	5	8
$F(x)$	0.10	0.30	0.45	0.65	0.75	0.90	0.95	1.00

(a) Obtain the probability distribution of X.
(b) Find $P(X$ is even$)$ and $P(1 \leq X < 8)$.
(c) Compute $P(X = -3 \mid X < 0)$ and $P(X \geq 3 \mid X > 0)$.
(d) Determine the mean and standard deviation of X.
(e) Calculate the mean and variance of $Y = X^2 - 3X$.

Ex 2.7. *In a gambling game a woman is paid $\$3$ if she draws a jack or a queen and $\$5$ if she draws a king or an ace from an ordinary deck of 52 playing cards. If she draws any other card, she loses. How much should she pay to play if the game is fair?*

Ex 2.8. *A couple decides to have 3 children. If none of the three is a girl, they will try again; and if they still don't get a girl, they will try once more. If a random variable X denotes the number of children the couple will have following this scheme, what is the expected value of X?*

Ex 2.9. *(Random Walk Problem). Starting from the origin, unit steps are taken to the right with probability p and to the left with probability q $(= 1-p)$. Assuming independent movements, find the mean and variance of the distance moved from the origin after n steps.*

Binomial Distribution

Ex 2.10. *If an unfair coin is flipped, it lands on head with probability 0.4. Suppose this coin is flipped 30 times successively. Find the probability that*
(a) it lands on heads exactly 8 times.
(b) it lands on tails at least 5 times.
(c) it lands on tails at most 20 times.

Ex 2.11. *If you buy a lottery ticket in 50 lotteries, in each of which your chance of winning a prize is $\frac{1}{100}$, what is the (approximate) probability that you will win a prize*
(a) at least once?
(b) exactly once?
(c) at least twice?

Ex 2.12. *A coffee connoisseur claims that he can distinguish between a cup of instant coffee and a cup of percolator coffee 75% of the time. It is agreed that his claim will be accepted if he correctly identifies at least 5 of the 6 cups. Find his chances of having the claim (a) accepted, (b) rejected, when he does have the ability he claims.*

Ex 2.13. *When coin* 1 *is flipped, it lands on heads with probability* 0.4*; when coin* 2 *is flipped, it lands on heads with probability* 0.7*. One of these coins is randomly chosen and flipped* 10 *times.*
(a) What is the probability that the coin lands on heads on exactly 7 *of the* 10 *flips?*
(b) Given that the first of these 10 *flips lands heads, what is the conditional probability that exactly* 7 *of the* 10 *flips land on heads?*

Ex 2.14. *A multiple choice test consists of* 8 *questions with* 3 *answers to each question (of which only one is correct). A student answers each question by rolling a regular die and checking the first answer if he gets* 1 *or* 2*, the second answer if he gets* 3 *or* 4 *and the third answer if he gets* 5 *or* 6*. To qualify the test, the student must secure at least* 75% *correct answers. If there is no negative marking, what is the probability that the student qualifies the test?*

Poisson Distribution

Ex 2.15. *The monthly worldwide average number of airplane crashes of commercial airlines is* 3.5*. What is the probability that there will be*
(a) at least 2 *such accidents in the next month,*
(b) at most 1 *accident in the next month?*

Ex 2.16. *In a certain factory turning out fountain pens, there is a small chance,* $\frac{1}{500}$*, for any pen to be defective. The pens are supplied in packets of* 10*. Calculate the approximate number of packets containing (i) one defective (ii) two defective pens in a consignment of* $20,000$ *packets.*

Ex 2.17. *The average number of incoming calls at a telephone booth between 10:00 A.M. and 11:00 A.M. is* 4.5 *per two minutes. Assume that the number of phone calls is a Poisson random variable. Determine the probability that the number of incoming calls at this booth during 10:15 A.M. to 10:25 A.M. is*
(a) 30 *or fewer.*
(b) more than 25*.*

Ex 2.18. *If* X *is a Poisson random variate such that* $P(X = 2) = \frac{2}{3}P(X = 1)$*, find* $P(X = 0)$ *and* $P(X = 5)$*.*

Ex 2.19. *A manager accepts the work submitted by his typist only when there is no mistake in the work. The typist has to type on an average* 20 *letters per day of about* 200 *words each. Find the chance of her making a mistake*
(a) if less than 1% *of the letters submitted by her are rejected.*
(b) if on 90% *days all the letters submitted by her are accepted.*

Uniform Distribution

Ex 2.20. *The average weight that is gained by a person over the winter months is uniformly distributed and it ranges from* 0 *to* 30 *lbs. What will be the probability of a person that he will gain between* 10 *and* 15 *lbs in the winter months?*

Ex 2.21. *A bus company schedules a bus every* 30 *minutes at a certain bus stop. A man comes to the stop at a random time. Let the random variable* X *count the number of minutes he has to wait for the next bus. Assume* X *has a uniform distribution over the interval* $(0, 30)$*. Compute the probability that he has to wait at least* 10 *minutes for the next bus.*

Ex 2.22. *A boy and a girl agree to meet at a certain park between* 4 *and* 5 *P.M. They agree that the one arriving first will wait* 10 *minutes for the other to arrive. Assuming that the arrival times are independent and uniformly distributed, find the probability that they meet.*

Ex 2.23. *If* X *has a uniform distribution in* $[0, 1]$*, find the distribution (probability density function) of* $-2 \log X$*.*

Normal Distribution

Ex 2.24. *In an engineering examination, a student is considered to have failed, secured second class, first class and distinction, according as he scores less than* 45%*, between* 45% *and* 60%*, between* 60% *and* 75% *and above* 75% *respectively. In a particular year,* 10% *of the students failed in the examination and* 5% *got distinction. Find the percentage of students who have got first class and second class (assume normal distribution of marks).âĂŃ*

Ex 2.25. *The mean IQ (intelligence quotient) of a large number of children of age* 14 *was* 100 *and the standard deviation* 16*. Assuming that the distribution was normal, find*
(a) What percentage of the children had IQ under 80*?*
(b) Between what limits the IQ's of the middle 40% *of the children lay?*
(c) What percentage of the children had IQ's within the range $\mu \pm 1.96\sigma$*, where* μ *and* σ *denote the mean and s.d.?*

Ex 2.26. *Assume the heights of soldiers are normally distributed and their mean height to be* 68.22 *inches with a variance of* 10.8 *square inches. How many soldiers in a regiment of* 1000 *would you expect to be over* 6 *feet tall?*

Ex 2.27. *The local authorities in a certain city install* $10,000$ *electrical lamps in the streets of the city. If these lamps have an average life of* $1,000$ *burning hours with a standard deviation of* 200 *hours, assuming normality, what number of lamps might be expected to fail*
(a) in the first 800 *burning hours?*
(b) between 800 *and* $1,200$ *burning hours?*
(c) After what period of burning hours would you expect that 10% *of the lamps would fail?*
(d) After what period of burning hours would you expect that 10% *of the lamps would be still burning?*

Ex 2.28. *If* Z *is a standard normal variate, find the value of* c *such that* $P(|Z| \leq c) = 0.95$*.*

Ex 2.29. *If X and Y are independent random variables following $N(8,2)$ and $N(12,4\sqrt{3})$ respectively, find the value of λ such that $P(2X-Y \leq 2\lambda) = P(X+2Y \geq \lambda)$, where $N(\mu,\sigma)$ denotes a normal distribution with mean μ and standard deviation σ.*

Moments and Moment Generating Function

Ex 2.30. *Let X be a discrete random variable having a Bernoulli distribution, i.e., the probability mass function, $p_X(x)$, of X is*

$$p_X(x) = \begin{cases} p, & \text{if } x = 1 \\ 1-p, & \text{if } x = 0 \\ 0, & \text{elsewhere} \end{cases}$$

where $p \in (0,1)$ is a constant. Derive the moment generating function of X, if it exists.

Ex 2.31. *A random variable X has probability function*

$$p(x) = \frac{1}{2^x}, \quad x = 1, 2, 3, \cdots$$

Obtain the moment generating function and hence the mean and variance.

Ex 2.32. *Find the moment-generating function of a Poisson random variable X. Hence, find the first and second moments of X about the origin.*

ANSWERS OF EXERCISES

Ex 2.1

$$(a)\ \int_1^\infty 3x^{-4}\,dx = 1 \implies \text{valid pdf}, \qquad (b)\ F(x) = \begin{cases} 0, & x \leq 1, \\ 1 - \dfrac{1}{x^3}, & x > 1 \end{cases}$$

$$(c)\ P(X > 4) = \frac{1}{64}, \qquad (d)\ f_Y(y) = \begin{cases} \dfrac{3}{y(\ln y)^4}, & y > e, \\ 0, & \text{otherwise} \end{cases}$$

Ex 2.2

$$(a)\ \int_0^1 6x(1-x)\,dx = 1 \implies \text{valid pdf}, \qquad (b)\ F(x) = \begin{cases} 0, & x < 0, \\ 3x^2 - 2x^3, & 0 \leq x \leq 1, \\ 1, & x > 1 \end{cases}$$

$$(c)\ P(X \leq 0.5 \mid 0.3 \leq X \leq 0.7) = \frac{71}{124}, \qquad (d)\ k = \frac{1}{2}$$

Ex 2.3

$$(b)\ P(X = -2) = \frac{1}{2}, \quad P(X = 0) = \frac{1}{4}, \quad P(X = 2) = \frac{1}{4}$$

$$(c)\ P(X = 1) = 0, \quad P(X \leq 1) = \frac{3}{4}, \quad P(X = 2) = \frac{1}{4}, \quad P(0 < X \leq 2) = \frac{1}{4}$$

Ex 2.4

$$(a)\ P(X=x)=\begin{cases}\frac{1}{9}, & x=\pm1,\pm2,\pm3,\\ \frac{1}{3}, & x=0\end{cases}$$

$$(b)\ F(x)=\begin{cases}0, & x<-3,\\ \frac{1}{9}, & -3\le x<-2,\\ \frac{2}{9}, & -2\le x<-1,\\ \frac{1}{3}, & -1\le x<0,\\ \frac{2}{3}, & 0\le x<1,\\ \frac{7}{9}, & 1\le x<2,\\ \frac{8}{9}, & 2\le x<3,\\ 1, & x\ge 3\end{cases}$$

$$(c)\ P(Y=18)=\frac{1}{9},\quad P(Y=4)=\frac{1}{3},\quad P(Y=3)=\frac{1}{9},\quad P(Y=9)=\frac{1}{9},\quad P(Y=5)=\frac{1}{9},\quad P(Y=14)=\frac{1}{9},\quad P(Y=31)=\frac{1}{9}$$

Ex 2.5

$$(a)\ 16k=1\Rightarrow k=\frac{1}{16},\qquad (b)\ f(x)=\begin{cases}\frac{1}{4}(x-1)^3, & 1<x<3,\\ 0, & \text{otherwise}\end{cases}$$

$$(c)\ E(X)=\frac{13}{5},\qquad V(X)=\frac{4}{75},\qquad (d)\ P(0.5<X\le 2.5)=\frac{81}{256}$$

Ex 2.6

(a)

x	-3	-1	0	1	2	3	5	8
$P(X=x)$	0.10	0.20	0.15	0.20	0.10	0.15	0.05	0.05

$$(b)\ P(X\ \text{even})=0.30,\qquad P(1\le X<8)=0.50$$

$$(c)\ P(X=-3\mid X<0)=\frac{1}{3},\qquad P(X\ge 3\mid X>0)=\frac{1}{3}$$

$$(d)\ E(X)=1,\qquad \sigma=\sqrt{8.1},\qquad (e)\ E(Y)=6.1,\qquad V(Y)=188.49$$

Ex 2.7

$$E(X)=\frac{16}{13},\qquad \text{Fair entry fee}=\frac{16}{13}$$

Ex 2.8

$$E(X)=\frac{219}{64}$$

Ex 2.9

$$E(S_n) = n(p-q), \qquad V(S_n) = 4npq$$

Ex 2.10

$$(a)\ \binom{30}{8}(0.4)^8(0.6)^{22}, \qquad (b)\ 1-\sum_{k=0}^{4}\binom{30}{k}(0.6)^k(0.4)^{30-k}, \qquad (c)\ \sum_{k=0}^{20}\binom{30}{k}(0.6)^k(0.4)^{30-k}$$

Ex 2.11

$$(a)\ 1-\left(\frac{99}{100}\right)^{50}, \qquad (b)\ 50\left(\frac{1}{100}\right)\left(\frac{99}{100}\right)^{49}, \qquad (c)\ 1-\left(\frac{99}{100}\right)^{50}-50\left(\frac{1}{100}\right)\left(\frac{99}{100}\right)^{49}$$

Ex 2.12

$$(a)\ \binom{6}{5}(0.75)^5(0.25)+(0.75)^6, \qquad (b)\ 1-\left[\binom{6}{5}(0.75)^5(0.25)+(0.75)^6\right]$$

Ex 2.13

$$(a)\ \frac{1}{2}\binom{10}{7}(0.4)^7(0.6)^3+\frac{1}{2}\binom{10}{7}(0.7)^7(0.3)^3, \qquad (b)\ \frac{\frac{1}{2}\binom{9}{7}(0.4)^6(0.6)^3+\frac{1}{2}\binom{9}{7}(0.7)^6(0.3)^3}{\frac{1}{2}(0.4)+\frac{1}{2}(0.7)}$$

Ex 2.14

$$P(X\geq 6)=\sum_{k=6}^{8}\binom{8}{k}\left(\frac{1}{3}\right)^k\left(\frac{2}{3}\right)^{8-k}$$

Ex 2.15

$$(a)\ 1-e^{-3.5}(1+3.5), \qquad (b)\ e^{-3.5}(1+3.5)$$

Ex 2.16

$$\lambda=0.02, \qquad (i)\ 20000\,e^{-0.02}(0.02)\approx 392, \qquad (ii)\ 20000\,e^{-0.02}\frac{(0.02)^2}{2}\approx 4$$

Ex 2.17

$$\lambda=22.5, \qquad (a)\ P(X\leq 30)=\sum_{k=0}^{30}e^{-22.5}\frac{22.5^k}{k!}, \qquad (b)\ P(X>25)=1-\sum_{k=0}^{25}e^{-22.5}\frac{22.5^k}{k!}$$

Ex 2.18

$$\frac{e^{-\lambda}\lambda^2/2}{e^{-\lambda}\lambda}=\frac{2}{3}\Rightarrow\lambda=\frac{4}{3}, \qquad P(X=0)=e^{-4/3}, \qquad P(X=5)=e^{-4/3}\frac{(4/3)^5}{5!}$$

Ex 2.19

$$(a)\ e^{-20\lambda} > 0.99 \Rightarrow \lambda < -\frac{\ln(0.99)}{20}, \quad (b)\ (e^{-20\lambda})^{0.9} = 0.9 \Rightarrow \lambda = -\frac{\ln(0.9)}{18}$$

Ex 2.20

$$P(10 < X < 15) = \frac{15-10}{30} = \frac{1}{6}$$

Ex 2.21

$$P(X \geq 10) = \frac{20}{30} = \frac{2}{3}$$

Ex 2.22

$$P(\text{meet}) = 1 - \frac{(50)^2}{60^2} = \frac{11}{36}$$

Ex 2.23

$$Y = -2\log X, \quad f_Y(y) = \frac{1}{2}e^{-y/2}, \qquad y > 0$$

Ex 2.24

$$P(X < 45) = 0.10, \qquad P(X > 75) = 0.05, \quad \text{First class} = 35\%, \quad \text{Second class} = 50\%$$

Ex 2.25

$$(a)\ P(Z < -1.25) \approx 0.1056, \quad (b)\ (91.62,\ 108.38), \quad (c)\ 95\%$$

Ex 2.26

$$\mu = 68.22, \quad \sigma = \sqrt{10.8} \approx 3.286, \quad P(X > 72) \approx 0.125, \quad \text{Expected soldiers} \approx 125$$

Ex 2.27

$$(a)\ 10000\,P(Z < -1) \approx 1587, \quad (b)\ 10000\,[P(Z < 1) - P(Z < -1)] \approx 6826$$

$$(c)\ x \approx 744, \quad (d)\ x \approx 1256$$

Ex 2.28

$$P(|Z| \leq c) = 0.95 \Rightarrow c = 1.96$$

Ex 2.29

$$2X - Y \sim N(4, 52), \quad X + 2Y \sim N(32, 196), \quad \frac{2\lambda - 4}{\sqrt{52}} = \frac{32 - \lambda}{14}, \quad \lambda = \frac{280}{21}$$

Ex 2.30

$$M_X(t) = E(e^{tX}) = (1-p) + pe^t$$

Ex 2.31

$$M_X(t) = \sum_{x=1}^{\infty} \left(\frac{e^t}{2}\right)^x = \frac{e^t}{2-e^t}, \qquad E(X) = 2, \qquad V(X) = 2$$

Ex 2.32

$$M_X(t) = \exp[\lambda(e^t - 1)], \qquad E(X) = \lambda, \qquad E(X^2) = \lambda + \lambda^2$$

Chapter 3

Bivariate Random Variables

- Joint Probability Mass Function and Density Function
- Joint Distribution Function
- Marginal Probability Distribution
- Conditional Probability Distribution
- Covariance
- Correlation Coefficient
- Regression Analysis (partial and multiple correlation)

Learning objectives

- To understand and compute joint, marginal, and conditional probability distributions of bivariate random variables.
- To evaluate and interpret the covariance and correlation coefficient for measuring relationships between two variables.
- To apply regression analysis, including partial and multiple correlation, for prediction and statistical interpretation of data.

Module 03, MAT3003//Probability, Statistics and Reliability
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3.1 Two Dimensional / Bivariate Random Variable

3.1.1 Definition

- Let S be a sample space associated with a random experiment E. Let X and Y be two random variables defined on S. Then the ordered pair (X, Y) is called a two-dimensional (bivariate) random variable.
- The value of (X, Y) at a point $s \in S$ is given by the ordered pair of real numbers $(X(s), Y(s)) = (x, y)$, where $X(s) = x$, $Y(s) = y$.

3.1.2 Two-Dimensional Discrete Random Variable

- If the possible values of (X, Y) are finite or countably infinite, then (X, Y) is called a two-dimensional discrete random variable.
- It may be represented as (x_i, y_j), where

$$i = 1, 2, 3, \ldots, m, \quad j = 1, 2, 3, \ldots, n.$$

Example

Consider tossing a coin twice.

- X = number of heads in first toss
- Y = number of heads in second toss

Sample space:

$$S = \{HH, HT, TH, TT\}$$

s	HH	HT	TH	TT
$X(s)$	1	1	0	0
$Y(s)$	1	0	1	0

- The range of (X, Y) is:

$$\{(1, 1), (1, 0), (0, 1), (0, 0)\}$$

- Hence (X, Y) is a discrete two-dimensional random variable.

3.2 JOINT PROBABILITY MASS DISTRIBUTION / PMF

The function $P(x, y)$ is a joint probability mass function if:

1. $P(x, y) \geq 0$ for all (x, y)
2. $\sum_x \sum_y P(x, y) = 1$
3. $P(X = x, Y = y) = P(x, y)$

3.2.1 Marginal Probability Functions

$$P_X(x) = \sum_y P(x,y), \quad P_Y(y) = \sum_x P(x,y)$$

3.2.2 Conditional Probability Functions

$$P_{X|Y}(x|y) = \frac{P(x,y)}{P_Y(y)}, \quad P_{Y|X}(y|x) = \frac{P(x,y)}{P_X(x)}$$

3.2.3 Independence

If X and Y are independent, then

$$P_{X|Y}(x|y) = P_X(x), \quad P_{Y|X}(y|x) = P_Y(y)$$

Hence,

$$P(x,y) = P_X(x) \cdot P_Y(y)$$

3.3 JOINT DISTRIBUTION FUNCTION

$$F_X(x) = P(X < x), \quad F_Y(y) = P(Y < y)$$

Joint distribution function:

$$F_{XY}(x,y) = P(X < x, Y < y)$$

3.3.1 Properties

1. $F_{XY}(-\infty, -\infty) = 0$
2. $F_{XY}(x,y) \leq 1$
3. $0 \leq F_{XY}(x,y) \leq 1$
4. $F_{XY}(x,y)$ is a non-decreasing function in x and y

Example

Consider the joint PMF given in tabular form.

$X \backslash Y$	0	1	2
0	0.25	0.10	0.15
1	0.14	0.35	0.01

Find the marginal PMF in x and y. Also find the conditional probability $P_{x/y}$ and $P_{y/x}$. Check whether x and y are independent or not.

$$P_X(x) = \sum_y P(x,y), \quad P_Y(y) = \sum_x P(x,y)$$

3.4 TWO-DIMENSIONAL CONTINUOUS RANDOM VARIABLE

- If (X, Y) can assume all values in a region R in the XY-plane, then (X, Y) is called a two-dimensional continuous random variable.

3.4.1 Joint Probability Mass Function

$$P(X = x_i, Y = y_j) = p_{ij}$$

3.5 Joint Density Function

For continuous random variables:

$$f(x, y) \geq 0, \quad \int\int f(x, y)\, dx\, dy = 1$$

Marginal densities:

$$f_X(x) = \int f(x, y)\, dy, \quad f_Y(y) = \int f(x, y)\, dx$$

Example

Find marginal distributions of X and Y. Also, find the conditional probability and check whether X and Y are independent or not for the following data:

$X\backslash Y$	1	2	3
1	0.10	0.05	0.05
2	0.15	0.20	0.05
3	0.05	0.15	0.20

Marginal distribution of X:

$$P(X = 1) = 0.20, \quad P(X = 2) = 0.40, \quad P(X = 3) = 0.40$$

Marginal distribution of Y:

$$P(Y = 1) = 0.30, \quad P(Y = 2) = 0.40, \quad P(Y = 3) = 0.30$$

Example

$X\backslash Y$	1	2	3	Total
1	0.05	0.01	0.01	0.07
2	0.05	0.01	0.01	0.07
3	0.40	0.25	0.21	0.86
Total	0.50	0.27	0.23	1

Marginal distribution of X:

$$P(X = 1) = 0.07, \quad P(X = 2) = 0.07, \quad P(X = 3) = 0.86$$

Marginal distribution of Y:

$$P(Y = 1) = 0.50, \quad P(Y = 2) = 0.27, \quad P(Y = 3) = 0.23$$

3.6 Covariance

If X and Y are two random variables, then the covariance between them is given by,

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E\big[(X-E(X))(Y-E(Y))\big] \\ &= E\big[XY - XE(Y) - YE(X) + E(X)E(Y)\big] \\ &= E(XY) - E(X)E(Y) - E(Y)E(X) + E(X)E(Y) \\ &= E(XY) - E(X)E(Y)\end{aligned}$$

If X and Y are independent random variables, then

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E(X)E(Y) - E(X)E(Y) \\ &= 0\end{aligned}$$

(Since if X and Y are independent, $E(XY) = E(X)E(Y)$)

Results

Result 1:

$$\mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X,Y)$$

Proof:

$$\begin{aligned}\mathrm{Cov}(aX,bY) &= E(aX\cdot bY) - E(aX)E(bY) \\ &= ab\,E(XY) - aE(X)\,bE(Y) \\ &= ab\big[E(XY) - E(X)E(Y)\big] \\ &= ab\,\mathrm{Cov}(X,Y)\end{aligned}$$

Result 2:

$$\mathrm{Cov}(X+a, Y+b) = \mathrm{Cov}(X,Y)$$

Proof:

$$\begin{aligned}\mathrm{Cov}(X+a,Y+b) &= E[(X+a)(Y+b)] - E(X+a)E(Y+b) \\ &= E[XY + bX + aY + ab] - [E(X)+a][E(Y)+b] \\ &= E(XY) + bE(X) + aE(Y) + ab \\ &\quad - [E(X)E(Y) + bE(X) + aE(Y) + ab] \\ &= E(XY) - E(X)E(Y) \\ &= \mathrm{Cov}(X,Y)\end{aligned}$$

Result 3:

$$\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$$

Proof:

$$\begin{aligned}
\operatorname{Cov}(aX+b, cY+d) &= E[(aX+b)(cY+d)] - E(aX+b)E(cY+d) \\
&= E[acXY + adX + bcY + bd] \\
&\quad - [aE(X)+b][cE(Y)+d] \\
&= acE(XY) + adE(X) + bcE(Y) + bd \\
&\quad - [acE(X)E(Y) + adE(X) + bcE(Y) + bd] \\
&= ac\big[E(XY) - E(X)E(Y)\big] \\
&= ac\operatorname{Cov}(X,Y)
\end{aligned}$$

Problem

The joint probability distribution of two random variables X and Y is given below:

$X \backslash Y$	1	2	Total
0	0.2	0.3	0.5
1	0.1	0.4	0.5
Total	0.3	0.7	1

Find $\operatorname{Cov}(X, Y)$.

Solution

Step 1: Compute $E(X)$

$$E(X) = \sum_x x \cdot P(X = x)$$

$$E(X) = 0(0.5) + 1(0.5) = 0.5$$

Step 2: Compute $E(Y)$

$$E(Y) = \sum_y y \cdot P(Y = y)$$

$$E(Y) = 1(0.3) + 2(0.7) = 0.3 + 1.4 = 1.7$$

Step 3: Compute $E(XY)$

$$E(XY) = \sum_x \sum_y xy \cdot P(X = x, Y = y)$$

$$E(XY) = (0 \cdot 1)(0.2) + (0 \cdot 2)(0.3) + (1 \cdot 1)(0.1) + (1 \cdot 2)(0.4)$$

$$E(XY) = 0 + 0 + 0.1 + 0.8 = 0.9$$

Step 4: Compute Covariance

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$\text{Cov}(X, Y) = 0.9 - (0.5)(1.7)$$

$$\text{Cov}(X, Y) = 0.9 - 0.85 = 0.05$$

Final Answer

$$\boxed{\text{Cov}(X, Y) = 0.05}$$

3.7 CORRELATION AND REGRESSION

A researcher conducted a study of the Aptitude test and Productivity after three months of training workers in a factory.

Aptitude test	1	3	5	7	9	2	4	6
Productivity	9	10	11	14	15	8	12	13

The researcher wants to determine whether there is a relationship between the Aptitude test and Productivity. It can be found from correlation.

3.8 Correlation

Correlation is the relationship that exists between two or more variables.

For Examples:

- Relationship between rainfall and wheat production.
- Relationship between price and demand.
- Relationship between insulin dose and blood sugar.

Types of Correlation

On the basis of direction:

- Positive
- Negative

On the basis of the ratio of change:

- Linear
- Non-linear

On the basis of number of variables:

- Simple
- Multiple
- Partial

Methods of Studying Correlation

- Scatter Diagram Method
- Karl PearsonâĂŹs Coefficient of Correlation
- SpearmanâĂŹs Rank Correlation

Scatter Diagram Method

- Data are plotted as dots on graph paper.
- Each pair (x, y) gives one point.
- The diagram obtained is called a scatter diagram.

3.9 Karl Pearson's Coefficient of Correlation

If X and Y are two variables, then:

$$r = \rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

Also,

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{\sum(x - \bar{x})^2 \sum(y - \bar{y})^2}}$$

Example

Calculate the coefficient of correlation between the marks obtained by 10 students:

Students	1	2	3	4	5	6	7	8	9	10
Accountancy	45	70	65	30	90	40	50	75	85	60
Statistics	35	90	70	40	95	40	60	80	80	50

Solution

$$\bar{X} = \frac{45 + 70 + 65 + 30 + 90 + 40 + 50 + 75 + 85 + 60}{10} = 61$$

$$\bar{Y} = \frac{35 + 90 + 70 + 40 + 95 + 40 + 60 + 80 + 80 + 50}{10} = 64$$

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \sum(Y - \bar{Y})^2}}$$

X	Y	$x = X - 61$	$y = Y - 64$	x^2	xy
45	35	-16	-29	256	464
70	90	9	26	81	234
65	70	4	6	16	24
30	40	-31	-24	961	744
90	95	29	31	841	899
40	40	-21	-24	441	504
50	60	-11	-4	121	44
75	80	14	16	196	224
85	80	24	16	576	384
60	50	-1	-14	1	14

$$\sum x^2 = 3490, \qquad \sum y^2 = 4380, \qquad \sum xy = 3535$$

$$r = \frac{3535}{\sqrt{3490 \times 4380}} \approx 0.904$$

$$\boxed{r \approx 0.904}$$

Properties of Correlation Coefficient

- $-1 \leq r \leq 1$
- Independent of change of origin and scale
- If the variables are independent, $r = 0$
- Based on mean and standard deviation

3.10 Spearman's Rank Correlation

SpearmanâĂŹs rank correlation is used when data are in ranks.

$$r = \rho(X, Y) = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

$$d = X - Y$$

Example

Calculate the rank correlation coefficient between the marks obtained by 10 students for the above example.

Merits and Limitations

Merits:

- Easy to compute
- Shows degree and direction of correlation

Limitations:

- Assumes linear relationship
- Difficult interpretation
- Affected by extreme values

3.11 Regression

According to Blair:

âĂIJRegression is the measure of the average relationship between two or more variables in terms of original units.âĂİ

Lines of Regression

- Regression line of Y on X:

$$y = a + bx$$

- Regression line of X on Y:

$$x = c + dy$$

3.12 Regression Equations

Regression of X on Y:

$$(X - \bar{X}) = r\frac{\sigma_X}{\sigma_Y}(Y - \bar{Y})$$

Regression of Y on X:

$$(Y - \bar{Y}) = r\frac{\sigma_Y}{\sigma_X}(X - \bar{X})$$

Example

From the following data, obtain the coefficient of correlation and both the regression equations:

X	2	4	6	8	10
Y	5	7	9	8	11

Solution

We first compute the means:

$$\bar{X} = \frac{2+4+6+8+10}{5} = 6$$

$$\bar{Y} = \frac{5+7+9+8+11}{5} = 8$$

Now calculate the deviations from the means.

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	x^2	xy
2	5	-4	-3	16	12
4	7	-2	-1	4	2
6	9	0	1	0	0
8	8	2	0	4	0
10	11	4	3	16	12
				$\sum x^2 = 40$	$\sum xy = 26$

Also,

$$\sum y^2 = (-3)^2 + (-1)^2 + 1^2 + 0^2 + 3^2 = 9 + 1 + 1 + 0 + 9 = 20$$

Hence, the coefficient of correlation is

$$r = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$$

$$r = \frac{26}{\sqrt{40 \times 20}} \approx 0.919$$

Therefore,

$$\boxed{r \approx 0.919}$$

Regression equation of Y on X

The regression coefficient is

$$b_{yx} = \frac{\sum xy}{\sum x^2} = \frac{26}{40} = 0.65$$

The regression equation is

$$Y - \bar{Y} = b_{yx}(X - \bar{X})$$

$$Y - 8 = 0.65(X - 6)$$

$$\boxed{Y = 0.65X + 4.1}$$

Regression equation of X on Y

The regression coefficient is

$$b_{xy} = \frac{\sum xy}{\sum y^2} = \frac{26}{20} = 1.3$$

Thus,

$$X - \bar{X} = b_{xy}(Y - \bar{Y})$$

$$X - 6 = 1.3(Y - 8)$$

$$\boxed{X = 1.3Y - 4.4}$$

Hence, the required regression equations are

$$\boxed{Y = 0.65X + 4.1}$$

and

$$\boxed{X = 1.3Y - 4.4}$$

with coefficient of correlation

$$\boxed{r \approx 0.919}$$

Exercise

1. **Joint Probability Distribution:**
   The joint probability distribution of X and Y is given below:

$X \backslash Y$	1	2
0	0.2	0.3
1	0.1	0.4

   Verify that it is a valid joint distribution.

2. **Marginal Distribution:**
   From the above distribution, find marginal distributions of X and Y.

3. **Conditional Probability:**
   Using the same table, find:

$$P(X = 1 \mid Y = 2), \quad P(Y = 1 \mid X = 0)$$

4. **Joint Density Function:**
   Let the joint density function be:

$$f(x, y) = kxy, \quad 0 < x < 2,\ 0 < y < 1$$

   Find k.

5. **Marginal Density:**
   For the above density, find marginal densities $f_X(x)$ and $f_Y(y)$.

6. **Covariance:**
   Given:
   $$E(X) = 2, \quad E(Y) = 3, \quad E(XY) = 8$$
   Find covariance of X and Y.

7. **Correlation Coefficient:**
   If $\sigma_X = 2$, $\sigma_Y = 4$ and $\text{Cov}(X, Y) = 6$, find correlation coefficient.

8. **Karl Pearson's Correlation:**
   Calculate correlation coefficient for the data:

X	1	2	3	4	5
Y	2	4	5	4	5

9. **Spearman Rank Correlation:**
   Compute rank correlation for:

Rank X	1	2	3	4	5
Rank Y	2	1	4	3	5

10. **Regression Analysis:**
    Given the following data:

X	2	4	6	8	10
Y	3	5	7	9	11

    Find:

    - Regression equation of Y on X
    - Regression equation of X on Y

Answers

1. It is a valid joint probability distribution.

2. $$P_X(0) = 0.5$$
   $$P_X(1) = 0.5$$
   $$P_Y(1) = 0.3$$
   $$P_Y(2) = 0.7$$

3.

$$P(X = 1 \mid Y = 2) = \frac{4}{7}$$
$$P(Y = 1 \mid X = 0) = 0.4$$

4.

$$k = 1$$

5.

$$f_X(x) = \frac{x}{2}, \quad 0 < x < 2$$
$$f_Y(y) = 2y, \quad 0 < y < 1$$

6.

$$\mathrm{Cov}(X, Y) = 2$$

7.

$$\rho = 0.75$$

8.

$$\bar{X} = 3$$
$$\bar{Y} = 4$$
$$r = 0.775$$

9.

$$\rho = 0.8$$

10. Regression equation of Y on X:

$$Y = X + 1$$

Regression equation of X on Y:

$$X = Y - 1$$

Chapter 4

Testing of Hypothesis

- Testing of Hypothesis – Introduction: Types of errors, Critical region, Procedure of testing hypothesis
- Large Sample Tests: Z-test for Single Proportion, Z-test for Difference of Proportions, Z-test for Single Mean, Z-test for Difference of Means
- Small Sample Tests: Student's t-test (single mean and difference of means), F-test (equality of variances), Chi-Square Test – Goodness of Fit, Chi-Square Test – Independence of Attributes

Learning objectives

- Understand the fundamental concepts of statistical hypothesis testing, including the formulation of null and alternative hypotheses.
- Identify the two types of errors (Type I and Type II) and appreciate their practical significance in decision making.
- Apply large sample Z-tests to draw inferences about population proportions and means.
- Apply small sample tests (t-test, F-test, and χ^2-test) when the sample size is small or the population variance is unknown.
- Use the chi-square test to assess goodness of fit and independence of categorical attributes.

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4.1 Introduction to Testing of Hypothesis

Statistical inference is the process of drawing conclusions about a population based on sample data. One of the two major branches of statistical inference is **Testing of Hypothesis**; the other is *estimation*.

In many practical situations, we wish to verify whether a claim (hypothesis) about a population parameter is supported or contradicted by the available sample evidence. For instance:

- Is the average life of a certain brand of bulb equal to 1000 hours?
- Does a new drug reduce blood pressure more effectively than the existing one?
- Is the proportion of defective items in a batch less than 5%?

The formal framework for answering such questions is called *hypothesis testing*.

4.1.1 Null and Alternative Hypothesis

Definition 1. The **null hypothesis**, denoted H_0, is a statement about a population parameter that is assumed to be true until the sample evidence proves otherwise. It typically represents the status quo or a claim of no effect/no difference.

Definition 2. The **alternative hypothesis**, denoted H_1 (or H_a), is the statement that contradicts H_0. It represents the conclusion we wish to support if there is sufficient evidence against H_0.

Example 5. A manufacturer claims that the mean weight of a packet of biscuits is 200 g. A consumer watchdog suspects the actual mean is different.

$$H_0 : \mu = 200\,\text{g} \quad \text{(manufacturer's claim)}$$
$$H_1 : \mu \neq 200\,\text{g} \quad \text{(two-tailed alternative)}$$

The alternative hypothesis may be **one-tailed** (left or right) or **two-tailed**:

- $H_1 : \mu \neq \mu_0$ – Two-tailed test
- $H_1 : \mu > \mu_0$ – Right-tailed (upper) test
- $H_1 : \mu < \mu_0$ – Left-tailed (lower) test

4.1.2 Types of Errors

When we make a decision based on sample evidence, two kinds of errors can occur.

Definition 3.

- **Type I Error (α):** Rejecting H_0 when it is actually *true*. The probability of committing a Type I error is called the **level of significance**, denoted α.
- **Type II Error (β):** Failing to reject H_0 when it is actually *false*. The probability of a Type II error is denoted β.
- **Power of the test** $= 1 - \beta$: the probability of correctly rejecting a false H_0.

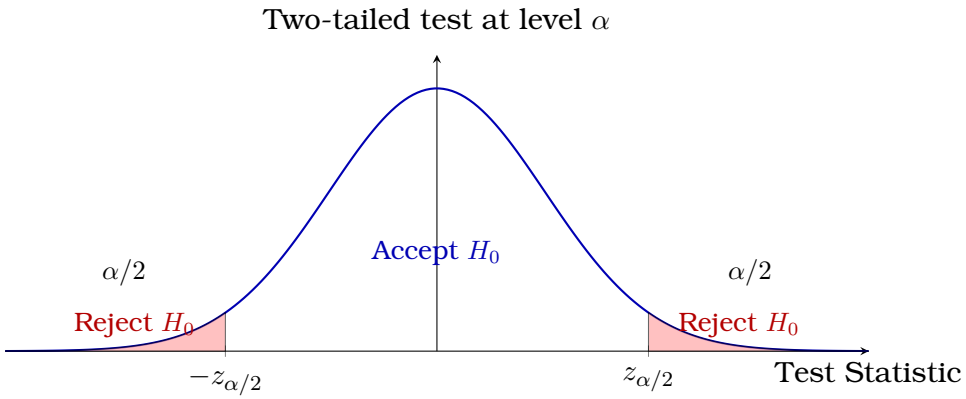
The relationship between the decision and the true state of nature is summarised in the following table:

	H_0 **is True**	H_0 **is False**
Reject H_0	Type I Error (α)	Correct Decision (Power $= 1 - \beta$)
Do not reject H_0	Correct Decision ($1 - \alpha$)	Type II Error (β)

Note 1. Reducing α (Type I error) generally increases β (Type II error) for a fixed sample size. Increasing the sample size is the most effective way to reduce both errors simultaneously.

4.1.3 Critical Region and Critical Value

Definition 4. The **critical region** (or rejection region) is the set of values of the test statistic for which H_0 is rejected. The boundary value(s) separating the critical region from the acceptance region is called the **critical value**.



Note 2. For a **two-tailed test** at level α, the critical values are $\pm z_{\alpha/2}$. Common values are: $z_{0.025} = 1.96$ for $\alpha = 0.05$ and $z_{0.005} = 2.576$ for $\alpha = 0.01$.

4.1.4 Level of Significance and p-value

Definition 5. The **level of significance** α is the maximum probability of committing a Type I error that the investigator is willing to tolerate. Common choices are $\alpha = 0.05$ (5%) and $\alpha = 0.01$ (1%).

Definition 6. The p**-value** is the probability of observing a test statistic as extreme as (or more extreme than) the one computed from the sample, *assuming* H_0 *is true*. A small p-value ($p < \alpha$) provides evidence against H_0.

4.1.5 General Procedure for Testing a Hypothesis

Procedure 1. The following steps constitute the standard procedure for testing a statistical hypothesis:

1. **State the hypotheses:** Formulate H_0 and H_1.

2. **Choose the level of significance** α**:** Typically 0.05 or 0.01.

3. **Identify the appropriate test statistic** and its distribution under H_0.

4. **Determine the critical region** (or compute the p-value).

5. **Compute the test statistic** from the sample data.

6. **Make a decision:**

   - If the test statistic falls in the critical region (or $p < \alpha$): *Reject* H_0.
   - Otherwise: *Fail to reject* H_0.

7. **State the conclusion** in the context of the problem.

4.1.6 Large Sample vs. Small Sample Tests

Feature	**Large Sample** ($n \geq 30$)	**Small Sample** ($n < 30$)
Test statistic	Z (standard normal)	t, F, or χ^2
Assumptions	CLT applies; σ may be unknown	Normal population; σ unknown
Distribution	$\mathcal{N}(0,1)$	t_{n-1}, $F_{(m-1,n-1)}$, χ^2_{k-1}

4.2 Large Sample Tests (Z-Tests)

When the sample size $n \geq 30$, by the Central Limit Theorem (CLT), the sampling distribution of the sample mean (and proportion) is approximately normal. The test statistic is therefore a Z-statistic.

4.2.1 Z-Test for a Single Proportion

Setting: A random sample of size n is drawn from a population. Let X be the number of elements possessing a certain attribute. The sample proportion is $\hat{p} = X/n$.

We wish to test $H_0 : p = p_0$ against a suitable alternative.

Test statistic:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\dfrac{p_0 q_0}{n}}}, \qquad \text{where } q_0 = 1 - p_0 \tag{4.1}$$

Under H_0, $Z \sim \mathcal{N}(0,1)$ for large n.

Critical values (at level α):

Alternative H_1	**Critical region**	**Critical value** ($\alpha = 0.05$)
$p \neq p_0$	$\lvert Z \rvert > z_{\alpha/2}$	1.96
$p > p_0$	$Z > z_\alpha$	1.645
$p < p_0$	$Z < -z_\alpha$	-1.645

Example 6. A manufacturer claims that at most 5% of his products are defective. A random sample of 400 items contains 24 defectives. Test the claim at $\alpha = 0.05$.

Solution.

$$n = 400, \quad X = 24, \quad \hat{p} = \frac{24}{400} = 0.06,$$
$$H_0 : p = 0.05, \quad H_1 : p > 0.05 \quad \text{(right-tailed test)}.$$

Test statistic:

$$Z = \frac{0.06 - 0.05}{\sqrt{\dfrac{0.05 \times 0.95}{400}}} = \frac{0.01}{\sqrt{0.00011875}} = \frac{0.01}{0.010898} \approx 0.917$$

Critical value: $z_{0.05} = 1.645$.

Since $Z = 0.917 < 1.645$, we **fail to reject** H_0. There is insufficient evidence at the 5% level to conclude that the proportion of defectives exceeds 5%.

4.2.2 Z-Test for Difference of Proportions

Setting: Two independent samples of sizes n_1 and n_2 with sample proportions $\hat{p}_1$ and $\hat{p}_2$. We test $H_0 : p_1 = p_2$.

Pooled estimate:

$$\hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}, \qquad \hat{q} = 1 - \hat{p} \tag{4.2}$$

Test statistic:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}} \tag{4.3}$$

Under H_0, $Z \sim \mathcal{N}(0, 1)$.

Example 7. In a survey, 480 out of 800 men and 340 out of 600 women are found to be literate. Test whether the literacy rates for men and women are significantly different at $\alpha = 0.01$.

Solution.

$$n_1 = 800, \ \hat{p}_1 = \frac{480}{800} = 0.60; \quad n_2 = 600, \ \hat{p}_2 = \frac{340}{600} = 0.567.$$
$$H_0 : p_1 = p_2, \quad H_1 : p_1 \neq p_2 \quad \text{(two-tailed)}.$$

Pooled proportion:

$$\hat{p} = \frac{800(0.60) + 600(0.567)}{800 + 600} = \frac{480 + 340}{1400} = \frac{820}{1400} \approx 0.5857, \quad \hat{q} = 0.4143.$$

Test statistic:

$$Z = \frac{0.60 - 0.567}{\sqrt{0.5857 \times 0.4143 \times \left(\frac{1}{800} + \frac{1}{600}\right)}} = \frac{0.033}{\sqrt{0.2427 \times 0.002917}} = \frac{0.033}{\sqrt{0.0007082}}$$
$$= \frac{0.033}{0.02661} \approx 1.240.$$

Critical value: $z_{0.005} = 2.576$.

Since $|Z| = 1.240 < 2.576$, we **fail to reject** H_0. The literacy rates are not significantly different at the 1% level.

4.2.3 Z-Test for a Single Mean

Setting: A large random sample of size n with sample mean $\bar{x}$ is drawn from a population with (known or estimated) standard deviation σ. We test $H_0 : \mu = \mu_0$.

Test statistic:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \tag{4.4}$$

If σ is unknown, replace it by the sample standard deviation s.

Example 8. The mean breaking strength of steel rods produced by a factory is claimed to be 60 kg. A sample of 100 rods has a mean breaking strength of 57.8 kg with standard deviation 10 kg. Test the claim at $\alpha = 0.05$.

Solution.

$$n = 100, \quad \bar{x} = 57.8, \quad s = 10,$$
$$H_0 : \mu = 60, \quad H_1 : \mu \neq 60 \quad \text{(two-tailed)}.$$

$$Z = \frac{57.8 - 60}{10/\sqrt{100}} = \frac{-2.2}{1} = -2.20.$$

Critical value: $z_{0.025} = 1.96$.

Since $|Z| = 2.20 > 1.96$, we **reject** H_0. The sample evidence suggests the mean breaking strength is significantly different from 60 kg at the 5% level.

4.2.4 Z-Test for Difference of Means

Setting: Two independent large samples of sizes n_1, n_2 with means $\bar{x}_1$, $\bar{x}_2$ and standard deviations σ_1, σ_2. We test $H_0 : \mu_1 = \mu_2$.

Test statistic:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}} \tag{4.5}$$

Example 9. The mean marks of 100 students from college A is 53.5 with $\sigma_1 = 8$, and the mean marks of 200 students from college B is 51.8 with $\sigma_2 = 7$. Test whether college A students perform significantly better at $\alpha = 0.05$.

Solution.

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 > \mu_2 \quad \text{(right-tailed)}.$$

$$Z = \frac{53.5 - 51.8}{\sqrt{\dfrac{64}{100} + \dfrac{49}{200}}} = \frac{1.7}{\sqrt{0.64 + 0.245}} = \frac{1.7}{\sqrt{0.885}} = \frac{1.7}{0.9409} \approx 1.807.$$

Critical value: $z_{0.05} = 1.645$.

Since $Z = 1.807 > 1.645$, we **reject** H_0. College A students perform significantly better at the 5% level.

4.3 Small Sample Tests

When the sample size is small ($n < 30$) and the population variance is unknown, the CLT approximation is inadequate. We then rely on exact distributions: the t-distribution, F-distribution, and chi-square distribution.

4.3.1 Student's t-Test

Assumptions: The population is normal; σ is unknown; sample size is small ($n < 30$). The test statistic follows a t-distribution with $\nu = n - 1$ degrees of freedom.

(a) t-Test for a Single Mean

We test $H_0 : \mu = \mu_0$ using

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \text{where} \quad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}. \tag{4.6}$$

The statistic follows t_{n-1} under H_0.

Example 10. Ten students are given intensive coaching and their marks in two tests (before and after coaching) are recorded. The scores *after* coaching are: 85, 76, 92, 65, 80, 74, 88, 70, 95, 78. Test whether the mean score is significantly greater than 75 at $\alpha = 0.05$.

Solution.

$$n = 10, \quad \bar{x} = \frac{85 + 76 + 92 + 65 + 80 + 74 + 88 + 70 + 95 + 78}{10} = \frac{803}{10} = 80.3.$$

Deviations $d_i = x_i - \bar{x}$: $4.7,\ -4.3,\ 11.7,\ -15.3,\ -0.3,\ -6.3,\ 7.7,\ -10.3,\ 14.7,\ -2.3$.

$$\sum d_i^2 = 22.09 + 18.49 + 136.89 + 234.09 + 0.09 + 39.69 + 59.29 + 106.09 + 216.09 + 5.29 = 838.10.$$

$$s = \sqrt{\frac{838.10}{9}} = \sqrt{93.122} \approx 9.650.$$

$$H_0 : \mu = 75, \quad H_1 : \mu > 75 \quad \text{(right-tailed)}.$$

$$t = \frac{80.3 - 75}{9.650/\sqrt{10}} = \frac{5.3}{3.051} \approx 1.737.$$

Tabulated $t_{9,\,0.05} = 1.833$ (one-tailed).

Since $t = 1.737 < 1.833$, we **fail to reject** H_0. The mean score is not significantly greater than 75 at the 5% level.

(b) Paired t-Test (Before-After or Matched Pairs)

When two related (paired) samples are compared, let $d_i = x_{1i} - x_{2i}$ be the difference for the i-th pair.

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad \nu = n - 1, \qquad \bar{d} = \frac{\sum d_i}{n}, \quad s_d = \sqrt{\frac{\sum d_i^2 - n\bar{d}^2}{n-1}}. \tag{4.7}$$

Example 11. The following data give the scores of 8 students before and after a training programme.

Student	1	2	3	4	5	6	7	8
Before (x_1)	45	52	61	48	50	54	63	57
After (x_2)	50	55	65	50	55	60	68	60

Test at $\alpha = 0.05$ whether the training has significantly improved the scores.

Solution. Differences $d_i = x_{2i} - x_{1i}$: $5, 3, 4, 2, 5, 6, 5, 3$.

$$\bar{d} = \frac{5+3+4+2+5+6+5+3}{8} = \frac{33}{8} = 4.125.$$

$$\sum d_i^2 = 25 + 9 + 16 + 4 + 25 + 36 + 25 + 9 = 149.$$

$$s_d = \sqrt{\frac{149 - 8(4.125)^2}{7}} = \sqrt{\frac{149 - 136.125}{7}} = \sqrt{\frac{12.875}{7}} = \sqrt{1.839} \approx 1.356.$$

$$H_0 : \mu_d = 0, \quad H_1 : \mu_d > 0 \quad \text{(right-tailed)}.$$

$$t = \frac{4.125}{1.356/\sqrt{8}} = \frac{4.125}{0.4796} \approx 8.601.$$

Tabulated $t_{7,\,0.05} = 1.895$.

Since $t = 8.601 \gg 1.895$, we **reject** H_0. The training has significantly improved the scores.

(c) Two-Sample t-Test (Independent Samples, Equal Variances)

For two independent samples of sizes n_1 and n_2 from normal populations with equal but unknown variances:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}, \quad \nu = n_1 + n_2 - 2, \tag{4.8}$$

where the **pooled standard deviation** is

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}. \tag{4.9}$$

Example 12. The tensile strength (in units) of two alloys is tested. Samples of sizes $n_1 = 8$ and $n_2 = 10$ give: $\bar{x}_1 = 52$, $s_1 = 4$, $\bar{x}_2 = 47$, $s_2 = 5$. Test at $\alpha = 0.05$ whether there is a significant difference in mean tensile strengths.

Solution.

$$H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2.$$

$$s_p = \sqrt{\frac{7(16) + 9(25)}{16}} = \sqrt{\frac{112 + 225}{16}} = \sqrt{\frac{337}{16}} = \sqrt{21.0625} \approx 4.589.$$

$$t = \frac{52 - 47}{4.589\sqrt{\frac{1}{8} + \frac{1}{10}}} = \frac{5}{4.589 \times 0.4743} = \frac{5}{2.177} \approx 2.297.$$

Tabulated $t_{16,\,0.025} = 2.120$ (two-tailed, $\alpha = 0.05$).

Since $t = 2.297 > 2.120$, we **reject** H_0. The difference in mean tensile strengths is significant.

4.3.2 F-Test for Equality of Variances

Definition 7. Let s_1^2 and s_2^2 be the sample variances of two independent samples of sizes n_1 and n_2 drawn from normal populations. The test statistic for $H_0 : \sigma_1^2 = \sigma_2^2$ is

$$F = \frac{s_1^2}{s_2^2} \quad \text{(take the larger variance in numerator).} \tag{4.10}$$

Under H_0, $F \sim F_{(n_1-1,\,n_2-1)}$.

Note 3. By convention we put the *larger* sample variance in the numerator, so $F \geq 1$, and we use only the upper critical value $F_{\alpha/2}(v_1, v_2)$ where $v_1 = n_1 - 1$, $v_2 = n_2 - 1$.

Example 13. Two random samples of sizes 10 and 12 give variances 28.8 and 16.2 respectively. Test at $\alpha = 0.05$ whether the population variances are equal.

Solution.

$$n_1 = 10,\ s_1^2 = 28.8; \quad n_2 = 12,\ s_2^2 = 16.2.$$

$$H_0 : \sigma_1^2 = \sigma_2^2, \quad H_1 : \sigma_1^2 \neq \sigma_2^2.$$

$$F = \frac{28.8}{16.2} \approx 1.778, \quad \nu_1 = 9, \quad \nu_2 = 11.$$

Tabulated $F_{0.025}(9, 11) \approx 3.59$.

Since $F = 1.778 < 3.59$, we **fail to reject** H_0. The population variances may be considered equal.

4.3.3 Chi-Square Test

The chi-square (χ^2) distribution arises naturally in the context of categorical data. Two important applications are the *goodness of fit* test and the *test of independence*.

(a) Chi-Square Test for Goodness of Fit

Purpose: To test whether an observed frequency distribution fits a hypothesised (theoretical) distribution.

Definition 8. Let O_i be the observed frequency and E_i the expected frequency for the i-th class. The test statistic is

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}, \tag{4.11}$$

which follows a chi-square distribution with $\nu = k - 1 - m$ degrees of freedom, where m is the number of population parameters estimated from the data.

Note 4. For the chi-square test to be valid, no expected frequency should be less than 5. If any $E_i < 5$, combine adjacent classes (called *pooling*).

Example 14. A die is rolled 120 times. The frequencies of faces 1 through 6 are 25, 17, 15, 23, 24, 16. Test at $\alpha = 0.05$ whether the die is unbiased.

Solution.

$$H_0 : \text{The die is unbiased (each face equally likely)},$$
$$H_1 : \text{The die is biased.}$$

Expected frequency for each face: $E_i = 120/6 = 20$.

Face	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2/E_i$
1	25	20	$+5$	1.25
2	17	20	-3	0.45
3	15	20	-5	1.25
4	23	20	$+3$	0.45
5	24	20	$+4$	0.80
6	16	20	-4	0.80
Total	120	120	0	**5.00**

$\chi^2_{\text{cal}} = 5.00$. Degrees of freedom: $\nu = 6 - 1 = 5$.
Tabulated $\chi^2_{5,\,0.05} = 11.07$.
Since $\chi^2_{\text{cal}} = 5.00 < 11.07$, we **fail to reject** H_0. The die appears to be unbiased.

Example 15. The number of defects per unit in a sample of 200 units from a production line is given below. Fit a Poisson distribution and test goodness of fit at $\alpha = 0.05$.

No. of defects (x)	0	1	2	3	4	≥ 5
Frequency (f)	94	62	30	10	3	1

Solution.

$$\bar{x} = \frac{0(94) + 1(62) + 2(30) + 3(10) + 4(3) + 5(1)}{200} = \frac{0 + 62 + 60 + 30 + 12 + 5}{200} = \frac{169}{200} = 0.845.$$

Estimated $\lambda = 0.845$. Poisson probabilities $P(X = x) = e^{-0.845}(0.845)^x/x!$:

x	$P(X = x)$	$E_i = 200\,P$	O_i
0	0.4296	85.92	94
1	0.3630	72.60	62
2	0.1534	30.68	30
3	0.0432	8.64	10
≥ 4	0.0108	2.16	4 (combined: $3 + 1$)

Combine last two rows (since $E_4 < 5$): $O = 4$, $E = 10.80$.

$$\chi^2 = \frac{(94-85.92)^2}{85.92} + \frac{(62-72.60)^2}{72.60} + \frac{(30-30.68)^2}{30.68} + \frac{(10-8.64)^2}{8.64} + \frac{(4-2.16)^2}{2.16}$$
$$= \frac{65.29}{85.92} + \frac{112.36}{72.60} + \frac{0.46}{30.68} + \frac{1.85}{8.64} + \frac{3.39}{2.16}$$
$$\approx 0.760 + 1.548 + 0.015 + 0.214 + 1.569 = 4.106.$$

Degrees of freedom: $\nu = 5 - 1 - 1 = 3$ ($m = 1$ parameter λ estimated).
Tabulated $\chi^2_{3,\,0.05} = 7.815$.
Since $\chi^2 = 4.106 < 7.815$, the Poisson distribution provides a **good fit**.

(b) Chi-Square Test for Independence of Attributes

Purpose: To test whether two categorical attributes A and B are statistically independent.

Contingency Table: Suppose attribute A has r categories and attribute B has c categories. The $r \times c$ contingency table has cell frequencies O_{ij}.

Expected frequency:

$$E_{ij} = \frac{(\text{Row total}_i) \times (\text{Column total}_j)}{\text{Grand total } N} \tag{4.12}$$

Test statistic:

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad \nu = (r-1)(c-1). \tag{4.13}$$

Example 16. A sample of 500 people is classified by gender and political preference as follows. Test at $\alpha = 0.01$ whether political preference is independent of gender.

	Party A	**Party B**	**Party C**	**Row Total**
Male	120	90	40	250
Female	100	95	55	250
Col. Total	220	185	95	500

Solution.

H_0 : Political preference is independent of gender.

Expected frequencies $E_{ij} = (\text{Row total}_i \times \text{Col total}_j)/500$:

$$E_{11} = \frac{250 \times 220}{500} = 110, \quad E_{12} = \frac{250 \times 185}{500} = 92.5, \quad E_{13} = \frac{250 \times 95}{500} = 47.5,$$
$$E_{21} = 110, \quad E_{22} = 92.5, \quad E_{23} = 47.5.$$

$$\chi^2 = \frac{(120-110)^2}{110} + \frac{(90-92.5)^2}{92.5} + \frac{(40-47.5)^2}{47.5}$$
$$+ \frac{(100-110)^2}{110} + \frac{(95-92.5)^2}{92.5} + \frac{(55-47.5)^2}{47.5}$$
$$= \frac{100}{110} + \frac{6.25}{92.5} + \frac{56.25}{47.5} + \frac{100}{110} + \frac{6.25}{92.5} + \frac{56.25}{47.5}$$
$$= 0.909 + 0.068 + 1.184 + 0.909 + 0.068 + 1.184 = 4.322.$$

Degrees of freedom: $\nu = (2-1)(3-1) = 2$.

Tabulated $\chi^2_{2,\,0.01} = 9.210$.

Since $\chi^2 = 4.322 < 9.210$, we **fail to reject** H_0. Political preference and gender are **not significantly associated** at the 1% level.

Example 17. In a 2×2 contingency table, the following data are observed:

	Attribute B_1	**Attribute** B_2	**Total**
A_1	30	70	100
A_2	50	50	100
Total	80	120	200

Test independence at $\alpha = 0.05$ using Yates' correction.

Solution.

For a 2×2 table with cells a, b, c, d and total N, Yates' corrected formula is:

$$\chi^2 = \frac{N\left(|ad - bc| - \dfrac{N}{2}\right)^2}{(a+b)(c+d)(a+c)(b+d)}. \tag{4.14}$$

Here $a = 30$, $b = 70$, $c = 50$, $d = 50$, $N = 200$, $ad = 1500$, $bc = 3500$.

$$\chi^2 = \frac{200\,(|1500 - 3500| - 100)^2}{100 \cdot 100 \cdot 80 \cdot 120} = \frac{200\,(1900)^2}{96{,}000{,}000} = \frac{200 \times 3{,}610{,}000}{96{,}000{,}000} = \frac{722{,}000{,}000}{96{,}000{,}000} \approx 7.52.$$

$\nu = 1$. Tabulated $\chi^2_{1,\,0.05} = 3.841$.

Since $\chi^2 = 7.52 > 3.841$, we **reject** H_0. Attributes A and B are **significantly associated**.

4.3.4 Summary of Critical Values

Test	**Statistic**	$\alpha = 0.05$	$\alpha = 0.01$
Z (two-tailed)	$Z \sim \mathcal{N}(0,1)$	± 1.96	± 2.576
Z (one-tailed)	$Z \sim \mathcal{N}(0,1)$	1.645	2.326
t ($\nu = 10$, two-tailed)	t_ν	2.228	3.169
χ^2 ($\nu = 5$)	χ^2_ν	11.07	15.09
F ($\nu_1 = 9, \nu_2 = 11$)	F_{ν_1,ν_2}	3.59	5.54

Exercises

Section 1: Large Sample Tests (Z-Tests)

1. A coin is tossed 400 times and heads appear 216 times. Is the coin biased? Test at $\alpha = 0.05$.

   [Answer: $Z = 1.60 < 1.96$; fail to reject H_0; coin is unbiased.]

2. In a city, 500 men are surveyed and 290 are found to own two-wheelers. In another city, 400 men are surveyed and 200 own two-wheelers. Test at $\alpha = 0.01$ whether the two cities differ significantly in two-wheeler ownership.

   [Answer: $\hat{p}_1 = 0.58$, $\hat{p}_2 = 0.50$, $\hat{p} = 0.544$; $Z \approx 2.47 > 2.576$; fail to reject at 1%, but reject at 5%.]

3. A sample of 64 observations has mean 47.5 and standard deviation 6. Test at $\alpha = 0.01$ whether the population mean is 50.

   [Answer: $Z = -3.33$; since $|Z| > 2.576$, reject H_0.]

4. The mean of a large sample of 1000 items is 80 units and standard deviation is 10 units. Test at $\alpha = 0.05$ whether the population mean is 85 units.

   [Answer: $Z = -15.81$; reject H_0.]

5. Two samples of sizes 400 and 500 have means 11.9 and 11.5 and standard deviations 4 and 3 respectively. Can the samples be regarded as drawn from the same population? Test at $\alpha = 0.05$.

   [Answer: $Z \approx 1.56 < 1.96$; fail to reject; samples may come from same population.]

Section 2: Student's t-Test

6. A random sample of 10 observations from a normal population: $14, 15, 12, 10, 11, 13, 9, 12, 14, 10$. Test whether the population mean is 11.5 at $\alpha = 0.05$.

   [Answer: $\bar{x} = 12$, $s \approx 1.944$; $t = 0.813 < t_{9,0.025} = 2.262$; fail to reject.]

7. The heights (in cm) of 8 randomly selected men are: 175, 178, 172, 168, 180, 177, 170, 174. Test at $\alpha = 0.05$ whether the mean height of men in the population is 170 cm.

   [Answer: $\bar{x} = 174.25$, $s \approx 3.77$; $t = 3.19 > t_{7,0.025} = 2.365$; reject H_0.]

8. Two independent samples of 7 and 6 observations have means 12.5 and 10.2 with $s_1^2 = 4.9$ and $s_2^2 = 5.8$. Test whether the means differ significantly at $\alpha = 0.05$.

   [Answer: $s_p \approx 2.295$; $t \approx 1.90$; $t_{11,0.025} = 2.201$; fail to reject.]

9. The following data give the blood pressure readings (mm Hg) of 6 patients before and after a treatment:

Patient	1	2	3	4	5	6
Before	140	148	135	152	143	160
After	128	135	130	140	130	148

   Test at $\alpha = 0.05$ whether the treatment has significantly reduced blood pressure.

   [Answer: $\bar{d} = 11.17$, $s_d \approx 3.06$; $t \approx 8.95 > t_{5,0.05} = 2.015$; reject H_0; treatment is effective.]

Section 3: F-Test

10. Two samples of sizes 13 and 11 give $s_1^2 = 24$ and $s_2^2 = 18$. Test at $\alpha = 0.05$ whether the variances are equal.

    [Answer: $F = 24/18 = 1.33$; $F_{0.025}(12, 10) \approx 3.62$; fail to reject.]

11. Two groups of students are tested. Group I ($n = 16$, $s^2 = 120$) and Group II ($n = 9$, $s^2 = 70$). Test at $\alpha = 0.05$ whether the populations have equal variances.

    [Answer: $F = 120/70 \approx 1.71$; $F_{0.025}(15, 8) \approx 4.10$; fail to reject.]

Section 4: Chi-Square Tests

12. The following table shows the results of a survey on preferences for three types of transport: Bus (80), Train (50), Car (70). Total = 200. Test at $\alpha = 0.05$ whether the preferences are equally distributed.

    [Answer: $E_i = 200/3 \approx 66.67$; $\chi^2 \approx 4.60$; $\chi^2_{2,0.05} = 5.991$; fail to reject.]

13. Fit a binomial distribution to the following data ($n = 6$) and test goodness of fit:

x	0	1	2	3	4	5	6	Total
f	5	18	28	12	7	6	4	80

    [Hint: Estimate $p = \bar{x}/n$. Pool tails to ensure all $E_i \geq 5$.]

14. The following 2×3 contingency table records the opinions of male and female respondents on a policy:

	Favour	**Indifferent**	**Oppose**	**Total**
Male	80	50	70	200
Female	60	40	100	200
Total	140	90	170	400

    Test at $\alpha = 0.05$ whether opinion is independent of gender.

    [Answer: $\chi^2 \approx 9.37 > \chi^2_{2,0.05} = 5.991$; reject H_0; opinion and gender are associated.]

15. In a sample of 400 persons, the data on smoking habit and lung disease is:

	Lung disease	**No lung disease**	**Total**
Smokers	60	140	200
Non-Smokers	20	180	200
Total	80	320	400

    Test at $\alpha = 0.01$ whether smoking and lung disease are associated. Use Yates' correction.

    [Answer: $\chi^2 \approx 18.05 > \chi^2_{1,0.01} = 6.635$; reject H_0; smoking and lung disease are significantly associated.]

Statistical Tables (Reference)

Table 1: Standard Normal (Z) Critical Values

Level of significance α	**One-tailed** z_α	**Two-tailed** $z_{\alpha/2}$
0.10	1.282	1.645
0.05	1.645	1.960
0.025	1.960	2.240
0.01	2.326	2.576
0.005	2.576	2.807

Table 2: t-Distribution Critical Values ($t_{\nu,\alpha}$, two-tailed)

ν	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.02$	$\alpha = 0.01$
1	6.314	12.706	31.821	63.657
5	2.015	2.571	3.365	4.032
8	1.860	2.306	2.896	3.355
10	1.812	2.228	2.764	3.169
15	1.753	2.131	2.602	2.947
20	1.725	2.086	2.528	2.845
25	1.708	2.060	2.485	2.787
∞	1.645	1.960	2.326	2.576

Table 3: Chi-Square Critical Values ($\chi^2_{\nu,\alpha}$)

ν	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.01$
1	2.706	3.841	5.024	6.635
2	4.605	5.991	7.378	9.210
3	6.251	7.815	9.348	11.345
4	7.779	9.488	11.143	13.277
5	9.236	11.070	12.833	15.086
10	15.987	18.307	20.483	23.209

Chapter 5

Design of Experiments and Reliability Engineering

- Analysis of Variance (ANOVA): One-Way Classification, Two-Way Classification
- Experimental Designs: Completely Randomized Design (CRD), Randomized Block Design (RBD)
- Reliability Engineering: Reliability Function, Hazard Function, Reliabilities of Series and Parallel Systems, System Reliability

Learning objectives

- Perform one-way and two-way ANOVA, construct ANOVA tables, and test hypotheses using the F-statistic.
- Apply Completely Randomized Design (CRD) and Randomized Block Design (RBD) to real-world problems.
- Evaluate reliability measures, including reliability and hazard functions, and compute reliability of series and parallel systems.

Module 05, MAT3003//Probability, Statistics and Reliability
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5.1 INTRODUCTION

Analysis of variance (ANOVA) is statistical technique used for analyzing the difference between the means of more than two samples. It is a parametric test of hypothesis. It is a step wise estimation procedures (such as the "variation" among and between groups) used to attest the equality between two or more population means.

ANOVA was developed by the statistician Ronald A. Fisher. Although several statisticians contributed to its development, it became widely recognized after its inclusion in Fisher's 1925 book Statistical Methods for Research Workers. The method is based on the principle of partitioning total variation, where the observed variance in a dataset is decomposed into components attributable to different sources of variation.

ANOVA provides a systematic approach for testing differences among group means and can be viewed as a generalization of the t-test when comparing more than two groups. The technique employs the F-test to determine whether the observed differences among sample means are statistically significant.

5.1.1 Assumptions

When applying one way analysis of variance there are three key assumptions that should be satisfied.

1. The observations are obtained independently and randomly from the populations defined by the factor levels.

2. The population at each factor level is (approximately) normally distributed.

3. These normal populations have a common variance (σ^2).

5.2 TESTING EQUALITY OF POPULATION (TREATMENT) MEANS: ONE-WAY ANOVA

In one-way classification, the following steps are carried out for computing the F-ratio through the most popular method:

1. Firstly, get the squared value of all the observations for different samples (columns).

2. Get the sum total of sample observations as $\sum X_1, \sum X_2, \ldots, \sum X_k$ in each column.

3. Get the sum total of squared values for each column as $\sum X_1^2, \sum X_2^2, \ldots, \sum X_k^2$ in each column.

4. Find the value of T by adding all the sums of sample observations, i.e.,

$$T = \sum X_1 + \sum X_2 + \cdots + \sum X_k$$

5. Compute the Correction Factor (C.F.) by the formula C.F. $= \frac{T^2}{n}$, where n is the number of observations.

6. Find the Total Sum of Squares (SST) using squared values and C.F.:

$$\text{SST} = \sum X_1^2 + \sum X_2^2 + \cdots + \sum X_k^2 - \text{C.F.}$$

7. Find the Sum of Squares between the samples (SSC) using:

$$\text{SSC} = \frac{(\sum X_1)^2}{n_1} + \frac{(\sum X_2)^2}{n_2} + \cdots + \frac{(\sum X_k)^2}{n_k} - \text{C.F.}$$

8. Finally, find the Sum of Squares within samples (SSE) as:

$$\text{SSE} = \text{SST} - \text{SSC}.$$

ANALYSIS OF VARIANCE (ANOVA) TABLE

Source of Variation	**Sum of squares**	**Degrees of Freedom**	**Mean square**	**Variance ratio of F**
Between samples	SSC	$v_1 = c - 1$ or $r - 1$	MSC=$\frac{SSC}{v_1}$	
Within samples	SSE	$v_2 = n - c$	MSE=$\frac{SSE}{v_2}$	F=$\frac{MSC}{MSE}$
Total	SST	$n - 1$		

SSC = Sum of squares between samples (columns)

SST = Total sum of squares of variations

SSE = Sum of squares within samples

MSC = Mean sum of squares between samples

MSE = Mean sum of squares within samples.

Illustration 1: To test the significance of variation in the retail prices of a commodity in three principal cities, Mumbai, Kolkata, and Delhi, four shops were chosen at random in each city and the prices observed in rupees were as follows:

City	**Shop 1**	**Shop 2**	**Shop 3**	**Shop 4**
Mumbai	15	7	11	13
Kolkata	14	10	10	6
Delhi	4	10	8	8

Do the data indicate that the price in the three cities are significantly different?

Solution: Let us take the null hypothesis that there is no significant difference in the prices of a commodity in the three cities.

Calculations for analysis of variance are as under:

Sample 1 (Mumbai)		**Sample 2 (Kolkata)**		**Sample 3 (Delhi)**	
x_1	x_1^2	x_2	x_2^2	x_3	x_3^2
15	225	14	196	4	16
7	49	10	100	10	100
11	121	10	100	8	64
13	169	6	36	8	64
$\sum x_1 = 46$	$\sum x_1^2 = 564$	$\sum x_2 = 40$	$\sum x_2^2 = 432$	$\sum x_3 = 30$	$\sum x_3^2 = 244$

There are $r = 3$ treatments (samples) with $n_1 = 4$, $n_2 = 4$, $n_3 = 4$, and $n = 12$.

$T =$ Sum of all the observations in the three samples

$$T = \sum x_1 + \sum x_2 + \sum x_3 = 46 + 40 + 30 = 116.$$

$$\text{CF (Correction Factor)} = \frac{T^2}{n} = \frac{(116)^2}{12} = 1121.33.$$

$$\text{SST (Total Sum of Squares)} = \left(\sum x_1^2 + \sum x_2^2 + \sum x_3^2\right) - \text{CF}$$
$$= (564 + 432 + 244) - 1121.33 = 118.67.$$

$$\text{SSC (Sum of Squares Between Samples)} = \left[\frac{(\sum x_1)^2}{n_1} + \frac{(\sum x_2)^2}{n_2} + \frac{(\sum x_3)^2}{n_3}\right] - \text{CF}$$

$$= \left[\frac{(46)^2}{4} + \frac{(40)^2}{4} + \frac{(30)^2}{4}\right] - 1121.33$$

$$= \left[\frac{2116}{4} + \frac{1600}{4} + \frac{900}{4}\right] - 1121.33$$

$$= \frac{4616}{4} - 1121.33 = 32.67.$$

$$\text{SSE (Error Sum of Squares)} = \text{SST} - \text{SSC} = 118.67 - 32.67 = 86.$$

Degrees of Freedom: $v_1 = r - 1 = 3 - 1 = 2, \quad v_2 = n - r = 12 - 3 = 9.$

$$\text{MSC} = \frac{\text{SSC}}{v_1} = \frac{32.67}{2} = 16.33$$

$$\text{MSE} = \frac{\text{SSE}}{v_2} = \frac{86}{9} = 9.55.$$

One-Way ANOVA TABLE

Source of Variation	**Sum of Squares**	**Degrees of Freedom**	**Mean Squares**	**Test-Statistic**
Between Samples	$32.67 = \text{SSTR}$	$2 = r - 1$	$\text{MSC} = \frac{32.67}{2} = 16.335$	$F = \frac{16.335}{9.55} = 1.71$
Within Samples	$86 = \text{SSE}$	$9 = n - r$	$\text{MSE} = \frac{86}{9} = 9.55$	
Total	$118.67 = \text{SST}$	$11 = n - 1$		

The table value of F for $v_1 = 2$, $v_2 = 9$, and $\alpha = 5\%$ level of significance is $F_{0.05} = 4.26$. Since the calculated value of F is less than the critical (table) value, the null hypothesis is accepted.

Hence, we conclude that the prices of a commodity in three cities have no significant difference.

5.3 TESTING EQUALITY OF POPULATION (TREATMENT) MEANS: TWO-WAY ANOVA

Two-Way ANOVA TABLE

Source of Variation	**Sum of squares**	**Degrees of Freedom**	**Mean square**	F**- ratio of** F
Between columns	SSC	$c-1$	MSC=$\frac{SSC}{c-1}$	$F_{\text{treatment}} = \frac{MSC}{MSE}$
Between rows	SSE	$r-1$	MSE=$\frac{SSE}{r-1}$	$F_{\text{block}} = \frac{MSR}{MSE}$
Residual error	SSE	$(c-1)(r-1)$	MSE=$\frac{SSE}{(c-1)(r-1)}$	
Total	SST	$n-1$ or $cr-1$		

Total variation consists of three parts: (i) variation between columns, SSC; (ii) variation between rows, SSR; and (iii) actual variation due to random error, SSE. That is,

SST = SSC + (SSR + SSE).

The degrees of freedom associated with SST are $cr-1$, where c and r are the number of columns and rows, respectively.

Degrees of freedom between columns : $df_c = c-1$

Degrees of freedom between rows : $df_r = r-1$

Degrees of freedom for residual error : $df_e = (c-1)(r-1)$.

The test statistic F for analysis of variance is given by:

$$F_{\text{treatment}} = \frac{MSC}{MSE}, \quad \text{use } \frac{MSC}{MSE} \text{ if } MSC > MSE, \text{ otherwise } \frac{MSE}{MSC}.$$

$$F_{\text{blocks}} = \frac{MSR}{MSE}, \quad \text{use } \frac{MSR}{MSE} \text{ if } MSR > MSE, \text{ otherwise } \frac{MSE}{MSR}.$$

Illustration 2: The following table gives the number of refrigerators sold by 4 salesmen in three months March, April and May:

Month	Salesman			
	A	B	C	D
March	50	40	48	39
April	46	48	50	45
May	39	44	40	39

Is there a significant difference in the sales made by the four salesmen?
Is there a significant difference in the sales made during different months?

Solution: Let us take the following null hypothesis:

H_0 : There is no significant difference in the sales made by the four salesmen.

H_0 : There is no significant difference in the sales made during different months.

The given data are coded by subtracting 40 from each observation. Calculations for a two-criteria (month and salesman) analysis of variance are shown below:

Month	Salesman								Row Sum
	A(x_1)	x_1^2	B(x_2)	x_2^2	C(x_3)	x_3^2	D(x_4)	x_4^2	
March	10	100	0	0	8	64	-1	1	17
April	6	36	8	64	10	100	5	25	29
May	-1	1	4	16	0	0	-1	1	2
Column Sum	15	137	12	80	18	164	3	27	48

$$\text{Row Sum or Column sum} = T = 48$$

$$\text{Correction Factor: (CF)} = \frac{T^2}{n} = \frac{48^2}{12} = 192$$

$$SSC = \left[\frac{15^2}{3} + \frac{12^2}{3} + \frac{18^2}{3} + \frac{3^2}{3}\right] - 192$$

$$= (75 + 48 + 108 + 3) - 192 = 42$$

$$SSR = \text{Sum of squares between months (rows)}$$

$$= \left[\frac{(17)^2}{4} + \frac{(29)^2}{4} + \frac{(2)^2}{4}\right] - 192$$

$$= (72.25 + 210.25 + 1) - 192 = 91.5$$

$$SST = \text{Total sum of squares}$$

$$= \left(\sum x_1^2 + \sum x_2^2 + \sum x_3^2 + \sum x_4^2\right) - \text{CF}$$

$$= (137 + 80 + 164 + 27) - 192 = 216$$

$$SSE = SST - (SSC + SSR) = 216 - (42 + 91.5) = 82.5$$

The total degrees of freedom are:

$$df = n - 1 = 12 - 1 = 11$$

$$df_c = c - 1 = 4 - 1 = 3, \quad df_r = r - 1 = 3 - 1 = 2, \quad df_e = (c-1)(r-1) = 3 \times 2 = 6$$

$$MSC = \frac{SSC}{c-1} = \frac{42}{3} = 14$$

$$MSR = \frac{SSR}{r-1} = \frac{91.5}{2} = 45.75$$

$$MSE = \frac{SSE}{(c-1)(r-1)} = \frac{82.5}{6} = 13.75$$

Two-Way ANOVA table

Source of variation	**Sum of squares**	**Degrees of freedom**	**Mean Squares**	**Variance Ratio**
Between Salesmen	$SSC = 42.0$	$c - 1 = 3$	$MSC = 14.00$	$F_{\text{treatment}} = 1.018$
Between months	$SSR = 91.5$	$r - 1 = 2$	$MSR = 45.75$	$F_{\text{block}} = 3.327$
Residual error	$SSE = 82.5$	$(c-1)(r-1) = 6$	$MSE = 13.75$	
Total	$SST = 216$	$n - 1 = 11$		

(a) The table value of $F = 4.75$ for $df_1 = 3$, $df_2 = 6$, and $\alpha = 5\%$. Since the calculated value of $F_{\text{treatment}} = 1.018$ is less than its table value, the null hypothesis is accepted. Hence we conclude that the sales made by the salesmen do not differ significantly.

(b) The table value of $F = 5.14$ for $df_1 = 2$, $df_2 = 6$, and $\alpha = 5\%$. Since the calculated value of $F_{\text{block}} = 3.327$ is less than its table value, the null hypothesis is accepted. Hence we conclude that sales made during different months do not differ significantly.

Illustration 3: A medical researcher wants to study the effect of type of prenatal care and age group of mothers on the birth weight of newborns (in kg). The following table shows the average birth weights (in kg):

Prenatal Care	**Below 25**	**Between 25-35**	**Above 35**
Regular Checkups	3.1	3.4	3.2
Minimal Checkups	2.8	3.0	2.9
No Checkups	2.5	2.7	2.6

Perform a Two-Way ANOVA to test:

(i) Whether prenatal care type has a significant effect on birth weight.

(ii) Whether mother's age group has a significant effect on birth weight. (Use significance level $\alpha = 0.05$).

Solution: Given data

Prenatal Care	Below 25	25–35	Above 35
Regular	3.1	3.4	3.2
Minimal	2.8	3.0	2.9
None	2.5	2.7	2.6

$T =$ total sum of observations $= 26.2, \quad n = 9.$

$$\text{Correction Factor (CF)} = \frac{T^2}{n} = 76.2711$$

Sum of Squares

$$SST = 0.8889, \quad SSR = 0.6022, \quad SSC = 0.0822.$$

$$SSE = TSS - SSR - SSC = 0.2045.$$

Degrees of Freedom

$$df = n - 1 = (rc - 1) = 8, \quad df_r = 2, \quad df_c = 2, \quad df_e = 4.$$

Mean Squares

$$MSR = 0.3011, \quad MSC = 0.0411, \quad MSE = 0.0511$$

Two-Way ANOVA table

Source	df	SS	MS	F-Ratio
Rows (Prenatal Care)	2	0.6022	0.3011	5.89
Columns (Age Group)	2	0.0822	0.0411	0.80
Error	4	0.2045	0.0511	—
Total	8	0.8889	—	—

F-Statistics

$$F_{\text{block}} = \frac{0.3011}{0.0511} = 5.89, \quad F_{\text{treatment}} = \frac{0.0411}{0.0511} = 0.80$$

Conclusion ($\alpha = 0.05$)

$$F_{0.5}(2, 4) = 6.94$$

$$F_{\text{block}} < F_{0.05}, \quad F_{\text{treatment}} < F_{0.05}$$

Hence, neither prenatal care nor age group has a significant effect on birth weight.

5.4 Exercise 1

1. Students were given different drug treatments before revising for their exams. Some were given a memory drug, some a placebo drug and some no treatment. The exam scores (%) are shown below for the three different groups:

	Memory Drug	Placebo	No Treatment
Score 1	70	37	3
Score 2	77	43	10
Score 3	83	50	17
Score 4	90	57	23
Score 5	97	63	30
Mean	83.40	50.00	16.60
Variance	112.30	109.00	112.30

   Carry out a one-way ANOVA to test the hypothesis that the treatments will have different effects.

   Answer:

$$F = 50.16, F_{0.05}(2, 12) = 3.89, F > F_{0.05}.$$

   There is a significant difference among the treatments. Hypothesis H_0 rejected.

2. In a comparison of the cleaning action of four detergents, 20 pieces of white cloth were first soiled with India ink. The cloths were then washed under controlled conditions with 5 pieces washed by each of the detergents. Unfortunately three pieces of cloth were 'lost' in the course of the experiment. Whiteness readings, made on the 17 remaining pieces of cloth, are shown below.

Detergent

A	B	C	D
77	74	73	76
81	66	78	85
61	58	57	77
76		69	64
69		63	

Assuming all whiteness readings to be normally distributed with common variance, test the hypothesis of no difference between the four brands as regards mean whiteness readings after washing.

Answer:

$$F = 1.50, F_{0.05}(3, 13) = 3.41, F < F_{0.05}.$$

There is no significant difference among the detergents. Hypothesis H_0 accepted.

3. A vehicle manufacturer wishes to test the ability of three types of steel-alloy panels to resist corrosion when three different paint types are applied. Three panels with differing steel-alloy composition are coated with three types of paint. The following coded data represent the ability of the painted panels to resist weathering.

Paint Type	**Steel-Alloy 1**	**Steel-Alloy 2**	**Steel-Alloy 3**
1	40	51	56
2	54	55	50
3	47	56	50

Use a two-way ANOVA procedure to determine whether any difference in the ability of the panels to resist corrosion may be assigned to either the type of paint or the steel-alloy composition of the panels.

Answer:

$$F_{\text{block}} = 0.25, F_{\text{treatment}} = 0.81, F_{0.05}(2, 4) = 6.94.$$

Neither paint type nor steel-alloy composition significantly affects corrosion resistance.

4. The following table gives per hectare yield for three varieties of wheat each grown on five plots:

Per hectare yield (in tons)			
Plot of land	**Varieties of wheat**		
	A	**B**	**C**
1	30	26	38
2	24	29	28
3	33	24	35
4	36	31	30
5	27	35	33

Construct a two-way ANOVA table and Test at 5% level of significance.

Answer:

$$F_{\text{treatment}} = 0.60, F_{\text{block}} = 0.35$$

$$F_{0.05}(2, 8) = 4.46, F_{0.05}(4, 8) = 3.84$$

There is no significant difference among wheat varieties or among plots.

5.5 DESIGN OF EXPERIMENTS

Design of Experiments (DOE) is defined as a branch of applied Statistics that deals with planning, conducting, analysing and interpreting controlled tests in an experiment to evaluate the factors that control the value of a parameter or group of parameters. We may think 'Design of Experiments' or an 'Experimental Design' as the laying out of a detailed experimental plan in advance of doing the experiment. It is a systematic method to determine the relationship between factors affecting a process and the output of that process. In other words, it is used to find cause-and-effect relationships in an experiment. This information is needed to manage process inputs in order to optimise the output. Experimental design can also be used at the point of greatest leverage to reduce design costs by speeding up the design process, reducing late engineering design changes, reducing product material and labor complexity. It involves not only the selection of suitable independent, dependent, and control variables, but planning the delivery of the experiment under statistically optimal conditions given the constraints of available resources. It allows for multiple input factors to be manipulated, determining their effect on a desired output (response). By manipulating multiple inputs at the same time, Design of Experiments can identify important interactions that may be missed when experimenting with one factor at a time.

5.5.1 COMPLETELY RANDOMISED DESIGN

The "Completely Randomised Design" (CRD) probably is the simplest type of design using two essential principles of replication and randomisation. All other Randomised designs stem from it by placing restrictions upon the allocation of the treatments within the experimental area. This design may be defined as one in which the treatments are randomly arranged over the whole of the experimental material such that every experimental unit has got the same chance of receiving a particular treatment. For the Completely Randomised Design, any difference among experimental units receiving the same treatment is considered as 'Experimental Error'. No effort is made to confine treatments to any portion of the entire area, material or space. Due to this reason, Completely Randomised Design is usually selected when the variation over all of the experimental material is relatively small. Completely Randomised Design is appropriate only for experiments with homogeneous experimental units, such as laboratory experiments, where environmental units are relatively easy to control. For field experiments, where there is generally large variation among experimental plots in such environmental factors as soil fertility, the Completely Randomised Design is rarely used.

Table 5.1: ANOVA Table for Completely Randomised Design

Source of Variation	**Degrees of Freedom**	**Sum of Squares**	**Mean Sum of Squares**	**F Ratio**
Between Treatments	$k-1$	SSTr	MSS_{Tr}	$F = \frac{\text{MSS}_{\text{Tr}}}{\text{MSS}_E}$
Error	$n-k$	SSE	MSS_E	df $(k-1,\ n-k)$
Total	$n-1$	TSS	—	—

Where $\dfrac{\text{SSTr}}{k-1} = \text{MSS}_{\text{Tr}}$, $\dfrac{\text{SSE}}{n-k} = \text{MSS}_E$.

Using the F distribution and the computed F Ratio, we test the null hypothesis of equality of Treatment effects (k effects; $k > 2$) and

(i) **Reject the Null Hypothesis** H_0**:** All the treatment effects are not significantly different, at the given level of significance α if the computed ratio,

$$F = \frac{\text{MSS}_{\text{Tr}}}{\text{MSS}_E} > F_{\alpha;(k-1),(n-k)},$$

the tabulated value;

where $F_{\alpha;(k-1),(n-k)}$ is the upper α point of the F distribution with degrees of freedom $(k-1,\, n-k)$, to be obtained from the F-table.

(ii) otherwise, **do not reject the Null Hypothesis** H_0, implying that all the treatment effects are equally effective.

Illustration 1: A farmer wishes to apply three types of fertilizers A, B and C, each replicated 10 times on 30 plots using Complete Randomisation process. All the 30 plots are divided into 5 blocks. Using the random numbers, find a layout of the Completely Randomised Design. If the yields per plot are as follows, carry out an analysis of the experiment and comment.

A	3.71	4.55	4.04	4.55	3.53	7.00	4.61	4.53	4.56	3.79
B	6.27	4.95	5.36	4.35	6.39	4.07	4.85	5.08	5.99	7.14
C	3.15	6.49	3.94	3.07	3.29	4.04	4.87	0.75	4.19	0.41

Solution: Given that the experimental area is divided into 5 blocks. So, each block will contain 6 plots. Since there are 3 treatments, each replicated 10 times, for allocation of treatments randomly on 30 plots we select 30 random numbers between 1 to 30 both inclusive.

The random numbers drawn are:

1, 4, 9, 27, 17, 20, 23, 24, 3, 30, 7, 11, 14, 29, 19, 22, 25, 2, 5, 8, 10, 13, 26, 18, 21, 12, 6, 16, 28, 15

Then, the Treatment A will be allotted first 10 random numbers, next 10 random numbers will be allotted to the Treatment B and last ten numbers will be allotted to Treatment C. The allocation of treatments to different plots along with their yields will be, therefore, as follows:

A 3.71	B 6.27	A 4.55	A 4.04	B 4.95	C 3.15
B 5.36	B 4.35	A 4.55	C 6.49	B 6.39	C 3.94
C 3.07	B 4.07	C 3.29	C 4.04	A 3.53	C 4.87
B 4.85	A 7.00	C 0.75	B 5.08	A 4.61	A 4.53
B 5.99	C 4.19	A 4.56	C 0.41	B 7.14	A 3.79

For the analysis of this data through the Analysis of Variance technique, we classify it into the form of one-way classified data as follows:

Treatment		
A	**B**	**C**
3.71	6.27	3.15
4.55	4.95	6.49
4.04	5.36	3.94
4.55	4.35	3.07
3.53	6.39	3.29
7.00	4.07	4.04
4.61	4.85	4.87
4.53	5.08	0.75
4.56	5.99	4.19
3.79	7.14	0.41

Each of these classes consists of 10 observations.

We observe that $n = 30$. The totals of each class are found to be:

$$T_1 = 44.87\ ; \quad T_2 = 54.45\ ; \quad T_3 = 34.20.$$

Therefore,

$$T_{..} = 133.52.$$

Therefore, Correction Factor,

$$\frac{T^2}{n} = \frac{(133.52)^2}{30} = 594.2530.$$

Raw Sum of Squares:

$$\sum_{i=1}^{k}\sum_{j=1}^{n} x_{ij}^2 = 661.0758.$$

$$\sum_{i=1}^{k} \frac{T_i^2}{n_i} = 201.3317 + 296.4802 + 116.9640 = 614.7759.$$

Hence, we have

$$\text{TSS} = 661.0758 - 594.2530 = 66.8228.$$

$$\text{SSTr} = 614.7759 - 594.2530 = 20.5229.$$

$$\text{SSE} = 66.8228 - 20.5229 = 46.2999.$$

The corresponding ANOVA table will be.

ANOVA Table for Completely Randomised Design

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Sum of Squares	F Ratio
Between Groups (Treatments)	2	20.5229	$MST = \frac{20.5229}{2} = 10.2614$	$F = \frac{10.2614}{1.7148}$
Error	27	46.2999	$MSE = \frac{46.2999}{27} = 1.7148$	$= 5.9840$
Total	29	66.8228	—	—

The calculated F Ratio comes out to be 5.984 for $(2, 27)$ degrees of freedom, whereas the tabulated value of F Ratio with $(2, 27)$ at 5% level of significance is 3.3541.

Since, $F_{\text{Calculated}} > F_{\text{Tabulated}}$, therefore, the null hypothesis is rejected indicating that the average yield under the treatments is not same.

5.5.2 RANDOMISED BLOCK DESIGN

Randomised Block Design (RBD) is an experimental design used when the experimental units are not completely homogeneous. In this design, the experimental units are first divided into groups called blocks such that the units within each block are relatively homogeneous. The treatments are then randomly assigned within each block. RBD is widely used in Agriculture Medical research, Biological experiments,Industrial experiments

The main purpose of blocking is to eliminate or reduce the effect of unwanted variation due to external factors. By doing so, the accuracy and reliability of the experimental results are improved.

5.5.3 Advantages of Randomised Block Design

The Randomised Block Design has many advantages over other designs. Due to this reason, Randomised Block Design is more frequently used than Completely Randomised Design. Some of the advantages of Randomised Block Design are listed below:

1. By means of grouping or blocking technique, a part of the predictable and unpredictable sampling variations is reduced to its minimum from the total experimental error. Therefore, more accurate results may be obtained by adopting Randomised Block Design.
2. It is a flexible design in the sense that any number of treatments and any number of replications may be used. If extra replication is needed for some treatments, this may be applied to more than one unit (but to the same number of units) per block.
3. This design utilises all the three principles of design, namely, Randomisation, Replication and Local Control. Hence, the conclusions drawn from Randomised Block Design are supposed to be more reliable and acceptable.
4. It also allows the use of different techniques of randomisation within different blocks. However, within a block, the technique of randomisation once adopted should remain the same.
5. It is easily adaptable. The analysis is simple and straightforward. Even if some data on blocks or treatments are missing, the analysis procedure is not very complicated.

5.5.4 Disadvantages of Randomised Block Design

1. We have seen that one of the most important requirements in this design is that the blocks should be internally homogeneous. Therefore, if blocks are not prepared with this condition, this design is the most disadvantageous due to the reason that a larger error term may creep up.
2. The design is applicable with an equal number of replications. However, if replications for each treatment are not the same, there is no option other than to use Completely Randomised Design.
3. If the number of treatments is very large, then the size of each block will increase whereas the number of blocks will decrease. This increases the sizes of the blocks, and, hence, may introduce heterogeneity within blocks.

ANOVA Table for Randomised Block Design

Source of Variation	Degrees of Freedom (df)	Sum of Squares (SS)	Mean Sum of Squares (MSS)	F – Ratio
Between Treatments	$(k-1)$	SST_r	$\frac{\text{SST}_r}{k-1} = \text{MSS}_{T_r}$	$F_T = \frac{\text{MSS}_{T_r}}{\text{MSS}_E}$
Between Blocks	$(r-1)$	SSB	$\frac{\text{SSB}}{r-1} = \text{MSS}_B$	$F_B = \frac{\text{MSS}_B}{\text{MSS}_E}$
Error	$(k-1)(r-1)$	SSE	$\frac{\text{SSE}}{(k-1)(r-1)} = \text{MSS}_E$	—-
Total	$(kr-1)$	SST	—-	—-

Illustration 2: The layout of a Randomised Block Design (RBD) with four blocks and five treatments (A, B, C, D, and E), along with the yields in each plot (measured in kilograms), is given below:

Layout and Yields per Plot

Block	Plot 1	Plot 2	Plot 3	Plot 4	Plot 5
Block I	B: 71	A: 78	C: 64	D: 72	E: 64
Block II	E: 63	A: 89	D: 70	C: 43	B: 72
Block III	C: 75	E: 70	B: 66	A: 68	D: 77
Block IV	A: 73	B: 68	C: 90	D: 70	E: 65

Construct a two-way classification table representing the yields categorized by Treatments (Rows) and Blocks (Columns).

solution Therefore, we classify the data into the form of a Two-way classified data as follows:

Block	Treatment				
	A	**B**	**C**	**D**	**E**
I	78	71	64	72	64
II	89	72	43	70	63
III	68	66	75	77	70
IV	73	68	90	70	65

Letting T_{iO} = Total of values in the ith Block ($i = 1, 2, 3, 4$); T_{Oj} = Total of values for the jth Treatment ($j = 1, 2, 3, 4, 5$) and $T = \sum_{i=1}^{4} T_{iO} = \sum_{j=1}^{5} T_{Oj}$.

We have:

$$T_{1O} = 349; \quad T_{2O} = 337; \quad T_{3O} = 356 \ \& \ T_{4O} = 366$$
$$T_{O1} = 308; \quad T_{O2} = 277; \quad T_{O3} = 272; \quad T_{O4} = 289; \quad T_{O5} = 262.$$

So that $T = 1408$.

Further, Raw Sum of Squares is $\sum_{i=1}^{4}\sum_{j=1}^{5} x_{ij}^2 = 100936$.

Sum $\sum_{i=1}^{4} \frac{T_{iO}^2}{5} = \frac{496062}{5} = 99212.4$.

$\sum_{j=1}^{5} \frac{T_{Oj}^2}{4} = \frac{397742}{4} = 99435.5$.

and Correction Factor $\frac{T^2}{20} = \frac{1982464}{20} = 99123.2$.

Therefore,

$$\begin{aligned}
\text{TSS} &= 100936 - 99123.2 = 1812.8 \\
\text{SSB} &= 99212.4 - 99123.2 = 89.2 \\
\text{SSTr} &= 99435.5 - 99123.2 = 312.3 \\
\text{SSE} &= 1812.8 - 89.2 - 312.3 = 1411.3.
\end{aligned}$$

ANOVA Table for Randomised Block Design

Source of Variation	**Degrees of Freedom (df)**	**Sum of Squares (SS)**	**Mean Sum of Squares (MSS)**	**F Ratio**
Between Treatment	$(5-1)$	$312.30 = \text{SST}_r$	$\frac{312.30}{4} = 78.075 = \text{MSST}_r$	$F_{T_r} = \frac{78.075}{117.61} = 0.664$
Between Blocks	$(4-1)$	$83.20 = \text{SSB}$	$\frac{89.20}{3} = 29.73 = \text{MSSB}$	$F_B = \frac{29.73}{117.61} = 0.253$
Error	$(5-1)(4-1)$	$1411.30 = \text{SSE}$	$\frac{1411.30}{12} = 117.61 = \text{MSSE}$	—-
Total	$(20-1)$	$1812.80 = \text{SST}$	—-	—-

Since the F-Ratio (Calculated) for Treatments is 0.664, which is found to be smaller than 3.26 (the F-Ratio Tabulated at $(4, 12)$ df and 5% level of significance observed from the F-table, the null hypothesis that the average yield of the vegetable under the 5 treatments are same is not rejected. This indicates that the effects of treatments on the yield of vegetable do not vary significantly.

Also, since the F-Ratio (Calculated) for blocks is 0.253, which is found to be smaller than 3.49 (the F-Ratio Tabulated at $(3, 12)$ df and 5% level of significance observed from the F-table, we do not reject the null hypothesis that the average yield of the vegetable does not significantly differ due to block effects.

Exercise 2

1. What do you mean by Design of Experiments? Explain the concepts of âĂIJCompletely Randomised DesignâĂİ.

2. In a field experiment, the effects of the five main Ammonium fertilizers on a vegetable were applied in a Completely Randomised Design (CRD) with the treatments A, B, C, D, and O. The plan and yield (in kg per plot) were observed as follows:

A	78	D	50	B	64	O	58	D	58	C	80
B	70	O	60	A	90	B	95	B	72	D	62
C	88	C	68	O	64	O	48	O	56	C	74
O	54	A	85	O	52	A	75	B	60	D	55

   Carry out an analysis of this experiment.

   Answer: At 5% level of significance, $F_{0.05}(4, 15) = 3.06$. Since $F_{\text{calculated}} = 3.35 > 3.06$, the treatment effects are significant. Hence, the ammonium fertilizers differ significantly in their effects on vegetable yield.

3. Explain the concepts of âĂIJRandomised Block DesignâĂİ. Discuss advantages and Disadvantages of Randomised Block Design.

4. In a field experiment, the effects of five main fertilizers A, B, C, D, and E on the yield of a vegetable was laid out in a Randomised Block Design. The plan and yields in kilogram per plot were found as follows:

Block I	B	71	A	78	C	64	D	72	E	64
Block II	E	63	A	89	D	70	C	43	B	72
Block III	C	75	E	70	B	66	A	68	D	77
Block IV	A	73	B	68	C	90	D	70	E	65

   Analyze the data mentioning the hypotheses to be tested and give your conclusions.

   Answer: At 5% level of significance, $F_{0.05}(4, 12) = 3.26$ and $F_{0.05}(3, 12) = 3.49$

   Since $F_{\text{treatment}} = 0.445 < 3.26$ and $F_{\text{block}} = 0.158 < 3.49$,

   both treatment and block effects are not significant.

   Hence, there is no significant difference among the fertilizers and no significant block effect.

5.6 RELIABILITY

Reliability theory is the branch of statistics and operations research devoted to studying how long systems will function properly. A "system" can refer to a single device, such as a DVR, or a network of devices or objects connected together (e.g., electronic components or stages in an assembly line). For any given system, the primary variable of interest is T = the system's lifetime, i.e., the duration of time until the system fails (either permanently or until repairs/upgrades are made). Since T measures time, we always have $T \geq 0$. Most often, T is modeled as a continuous random variable on $(0, \infty)$, though occasionally lifetimes are modeled as discrete or, at least, having positive probability of equaling zero (such as a light bulb that never turns on). The probability distribution of T is often described in terms of its reliability function.

5.6.1 Reliability Function

Let T denote the lifetime (i.e., the time to failure) of some system. The **reliability function** of T (or of the system), denoted by $R(t)$, is defined for $t \geq 0$ by

$$R(t) = P(T > t) = 1 - F(t) = \int_t^\infty f(t)\, dt, \tag{5.1}$$

where $F(t)$ is the cdf of T and $f(t)$ is a pdf of T. That is, $R(t)$ is the probability that the system lasts more than t time units. The reliability function is sometimes also called the *survival function* of T.

Properties of $F(t)$ and the relation $R(t) = 1 - F(t)$ imply that

1. If T is a continuous random variable on $[0, \infty)$, then $R(0) = 1$.
2. $R(t)$ is a non-increasing function of t.
3. $R(t) \to 0$ as $t \to \infty$.

5.6.2 Series and Parallel Designs

Suppose a system consists of n independent components with reliability functions $R_1(t), \ldots, R_n(t)$.

1. If the n components are connected in series, the system reliability function is

$$R(t) = \prod_{i=1}^{n} R_i(t). \tag{5.2}$$

2. If the n components are connected in parallel, the system reliability function is

$$R(t) = 1 - \prod_{i=1}^{n} [1 - R_i(t)]. \tag{5.3}$$

5.6.3 Mean Time to Failure (MTTF)

Suppose a system has reliability function $R(t)$ for $t \geq 0$. Then the system's mean time to failure is given by

$$\mu_T = \int_0^\infty [1 - F(t)]dt = \int_0^\infty R(t)dt. \tag{5.4}$$

5.6.4 Hazard Function

Let T denote the lifetime of a system. If the random variable T has probability density function $f(t)$ and cdf $F(t)$, the **hazard function** is defined by

$$h(t) = \frac{f(t)}{1 - F(t)} = \frac{f(t)}{R(t)}. \tag{5.5}$$

NOTE: Suppose a system has a continuous lifetime distribution on $[0, \infty)$ with hazard function $h(t)$. Then its lifetime (aka time to failure) has reliability function $R(t)$ given by

$$R(t) = e^{-\int_0^t h(u)du}.$$

and pdf $f(t)$ given by

$$f(t) = -R'(t) = h(t)e^{-\int_0^t h(u)du}.$$

Illustration 1: The failure density function (pdf) of the random variable T is given by:

$$f(t) = \begin{cases} 0.012e^{-0.012t}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

calculate the followings:

1. Reliability of the component, $R(t)$.
2. Reliability for a 100-hour mission time, $R(100)$.
3. Mean Time to Failure (MTTF).
4. Hazard rate (hazard function), $h(t)$.

Solution:

1. The reliability function is defined as $R(t) = 1 - F(t) = \int_t^\infty f(u)du$:

$$R(t) = \int_t^\infty 0.012e^{-0.012u}du = \left[-e^{-0.012u}\right]_t^\infty = 0 - (-e^{-0.012t}) = e^{-0.012t}.$$

2. Substitute $t = 100$ into the reliability function:

$$R(100) = e^{-0.012(100)} = e^{-1.2} \approx 0.3012$$

3. The MTTF is given by $\int_0^\infty R(t)dt$:

$$\text{MTTF} = \int_0^\infty e^{-0.012t}dt = \left[\frac{e^{-0.012t}}{-0.012}\right]_0^\infty = \frac{1}{0.012} \approx 83.33 \text{ hours}$$

4. The hazard function is defined as $h(t) = \frac{f(t)}{R(t)}$:

$$h(t) = \frac{0.012e^{-0.012t}}{e^{-0.012t}} = 0.012$$

Since the hazard rate is constant, this component follows an exponential distribution.

Illustration 2: The hazard rate of a component is given by $h(t) = 0.4 \times 10^{-4}t$, where t is in years. Calculate the reliability of the component for the first 5 years.

Solution: The relationship between the reliability function $R(t)$ and the hazard rate $\lambda(t)$ is given by the formula:

$$R(t) = \exp\left(-\int_0^t h(u)du\right)$$

Given $h(t) = 0.4 \times 10^{-4}t$, we substitute this into the integral:

$$R(t) = \exp\left(-\int_0^t 0.4 \times 10^{-4}u\,du\right)$$

Performing the integration, we get

$$\begin{aligned} R(t) &= \exp\left(-0.4 \times 10^{-4}\left[\frac{u^2}{2}\right]_0^t\right) \\ &= \exp\left(-0.2 \times 10^{-4}t^2\right). \end{aligned}$$

To find the reliability for the first 5 years, we substitute $t = 5$:

$$R(5) = e^{-0.2\times10^{-4}\cdot(5)^2}$$

$$R(5) = e^{-5.0\times10^{-4}}.$$

Illustration 3: A system consists of three components in series. The reliabilities of the components are $R_1 = 0.9$, $R_2 = 0.8$, and $R_3 = 0.7$. Assuming the components fail independently, what is the reliability of the system?

Solution: For a system where components are connected in **series**, the system reliability R_s is the product of the individual component reliabilities:

$$R_s = \prod_{i=1}^{n} R_i = R_1 \times R_2 \times R_3.$$

On substituting the values $R_1 = 0.9, R_2 = 0.8, R_3 = 0.7$, we get

$$R_s = (0.9) \times (0.8) \times (0.7)$$

$$R_s = 0.504$$

The reliability of the system is **0.504** (or 50.4%).

Illustration 4: A system has two subsystems, A and B, connected in parallel. Subsystem A has a reliability of 0.9, and subsystem B has a reliability of 0.8. What is the probability that the system will function?

Solution: For a system where components are connected in **parallel**, the system reliability $R(t)$ is

$$R(t) = 1 - \prod_{i=1}^{n}[1 - R_i(t)]$$

For two subsystems, A and B, the formula is:

$$R_s = 1 - [(1 - R_A) \times (1 - R_B)]$$

$$R_s = 1 - [(1 - 0.9) \times (1 - 0.8)]$$

$$R_s = 1 - [0.1 \times 0.2]$$

$$R_s = 1 - 0.02$$

$$R_s = 0.98.$$

The probability that the system will function (system reliability) is **0.98** (or 98%).

Illustration 5: A critical control system in a nuclear power plant uses a 4-out-of-6 configuration. Each of the 6 control units has a reliability of 0.95. What is the probability that the control system will fail (i.e., less than 4 units are functioning)?

Solution: This is a binomial reliability problem where: $n = 6$ (total number of units), $k = 4$ (minimum units required for the system to function), $p = 0.95$ (reliability of a single unit), $q = 1 - p = 0.05$ (probability of failure for a single unit).

The system functions if $4, 5,$ or 6 units are working. The probability of exactly x units functioning is given by the binomial distribution:

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

The system reliability is:

$$R_s = P(X \geq 4) = P(X = 4) + P(X = 5) + P(X = 6)$$

$$R_s = \binom{6}{4}(0.95)^4(0.05)^2 + \binom{6}{5}(0.95)^5(0.05)^1 + \binom{6}{6}(0.95)^6(0.05)^0$$

$$R_s \approx 0.03054 + 0.23213 + 0.73509 = 0.99776.$$

The probability of failure is:

$$P_{fail} = 1 - R_s \approx 1 - 0.99776 = 0.00224.$$

The probability that the control system will fail is approximately **0.00224** (or 0.224%).

Illustration 6: A certain machine has the following hazard function:

$$h(t) = \begin{cases} .002 & 0 < t \leq 200 \\ .001 & t > 200 \end{cases}$$

(a) Determine the reliability function.
(b) Determine the probability density function of the machineâĂŹs lifetime.
(c) Find the mean time to failure.

Solution:

(a) Reliability Function $R(t)$ is related to the hazard function by the relation $R(t) = e^{-\int_0^t h(u)du}$.

Case 1: $0 < t \leq 200$

$$R(t) = e^{-\int_0^t 0.002du} = e^{-0.002t}$$

Case 2: $t > 200$

$$R(t) = e^{-\left(\int_0^{200} 0.002du + \int_{200}^t 0.001du\right)}$$

$$R(t) = e^{-(0.4+0.001(t-200))} = e^{-(0.001t+0.2)} = e^{-0.2}e^{-0.001t}$$

Then, required reliability function $R(t)$:

$$R(t) = \begin{cases} e^{-0.002t} & 0 < t \leq 200 \\ e^{-(0.001t+0.2)} & t > 200. \end{cases}$$

(b) Probability Density Function $f(t)$ is given by $f(t) = h(t)R(t)$:

$$f(t) = \begin{cases} 0.002e^{-0.002t} & 0 < t \leq 200 \\ 0.001e^{-(0.001t+0.2)} & t > 200. \end{cases}$$

(c) Mean Time to Failure (MTTF) is the integral of the reliability function from 0 to infinity:

$$\text{MTTF} = \int_0^{200} e^{-0.002t}dt + \int_{200}^{\infty} e^{-(0.001t+0.2)}dt$$

$$\text{MTTF} = \left[\frac{e^{-0.002t}}{-0.002}\right]_0^{200} + e^{-0.2}\left[\frac{e^{-0.001t}}{-0.001}\right]_{200}^{\infty}$$

$$\text{MTTF} = \frac{1 - e^{-0.4}}{0.002} + \frac{e^{-0.2}e^{-0.2}}{0.001} = \frac{1 - 0.6703}{0.002} + \frac{0.6703}{0.001}$$

$$\text{MTTF} = 164.85 + 670.3 = 835.15 \text{ hours}.$$

Exercise 3

1. The lifetime T (in hours) of a specific electronic controller has a probability density function (pdf) given by:

$$f(t) = \begin{cases} 0.005e^{-0.005t}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Calculate the following:

(a) The reliability function $R(t)$.

(b) The probability that the controller survives at least 200 hours.

(c) The Mean Time to Failure (MTTF).

(d) The hazard function $h(t)$ and explain if the failure rate is increasing, decreasing, or constant.

Answer: (a) $R(t) = e^{-0.005t}$, (b) $P(T \geq 200) = e^{-1} \approx 0.3679$, (c) MTTF $= 200$ hours, (d) $h(t) = 0.005$. Failure rate is constant.

2. The failure density function of a system is given by:

$$f(t) = \frac{t}{50} e^{-t^2/100} \quad \text{for } t \geq 0$$

(where t is in hours).

(a) Find the Reliability function $R(t)$ for this system.

(b) Calculate the Mean Time To Failure (MTTF) using the derived Reliability function.

Answer: (a) $R(t) = e^{-t^2/100}$, (b) MTTF $= 5\sqrt{\pi} \approx 8.86$ hours

3. Suppose the hazard function of a device is given by

$$h(t) = \begin{cases} \alpha\left(1 - \dfrac{t}{\beta}\right) & 0 \leq t \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

for some $\alpha, \beta > 0$. This model asserts that if a device lasts β hours, it will last forever (while seemingly unreasonable, this model can be used to study just "initial wearout").

(a) Find the reliability function.

(b) Find the pdf of device lifetime.

Answer: (a) The Reliability Function $R(t)$ is given by $R(t) = \exp\left(-\int_0^t h(u)\,du\right)$.

For $0 \leq t \leq \beta$:

$$\int_0^t \alpha\left(1 - \frac{u}{\beta}\right) du = \alpha\left[u - \frac{u^2}{2\beta}\right]_0^t = \alpha\left(t - \frac{t^2}{2\beta}\right)$$

Thus, the reliability function is:

$$R(t) = \begin{cases} \exp\left(-\alpha t + \frac{\alpha t^2}{2\beta}\right) & 0 \leq t \leq \beta \\ \exp\left(-\frac{\alpha\beta}{2}\right) & t > \beta \end{cases}$$

(b) The Probability Density Function $f(t)$ is defined as $f(t) = h(t) \cdot R(t)$.

For $0 \leq t \leq \beta$:

$$f(t) = \alpha\left(1 - \frac{t}{\beta}\right) \exp\left(-\alpha t + \frac{\alpha t^2}{2\beta}\right)$$

For $t > \beta$, $f(t) = 0$.

4. A satellite control system uses a 4-out-of-7 configuration. Each unit has reliability 0.97. Find the probability that fewer than 4 units are functioning.

   Answer:

$$P(X < 4) = \sum_{k=0}^{3} \binom{7}{k} (0.97)^k (0.03)^{7-k} \approx 2.64 \times 10^{-5}.$$

5. A nuclear monitoring system uses a 5-out-of-6 configuration, where each unit has reliability 0.94. What is the probability that the system fails?

   Answer:

$$P(\text{Fail}) = P(X < 5) = 1 - [P(X = 5) + P(X = 6)] = 1 - 0.95407 = 0.04593.$$